

Rise of the Machines: Algorithmic Trading in the Foreign Exchange Market

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July 5, 2013

Abstract

We study the impact of algorithmic trading in the foreign exchange market using a long time series of high-frequency data that specifically identifies computer-generated trading activity. Using both a reduced-form and a structural estimation, we find clear evidence that algorithmic trading causes an improvement in two measures of price efficiency in this market: the frequency of triangular arbitrage opportunities and the autocorrelation of high-frequency returns. Relating our results to the recent theoretical literature on the subject, we show that the reduction in arbitrage opportunities is associated primarily with computers taking liquidity, while the reduction in the autocorrelation of returns owes more to the algorithmic provision of liquidity. We also find evidence that algorithmic traders do not trade with each other as much as a random matching model would predict, which we view as consistent with their trading strategies being highly correlated. However, the analysis shows that this high degree of correlation does not appear to cause a degradation in market quality.

*JEL Classification: F3, G12, G14, G15. Keywords: Algorithmic trading; Price discovery. Chaboud and Vega are with the Division of International Finance, Federal Reserve Board, Mail Stop 43, Washington, DC 20551, USA; Chiquoine is with the Stanford Management Company, 635 Knight Way, Stanford CA 94305, USA; Hjalmarsson is with University of Gothenburg, Department of Economics, Centre for Finance, Vasagatan 1, SE 405 30 Gothenburg, Sweden, and Queen Mary, University of London, School of Economics and Finance, Mile End Road, London E1 4NS, UK. Please address comments to the authors via e-mail at alain.p.chaboud@frb.gov, bchiquoine@tiff.org, e.hjalmarsson@qmul.ac.uk, and clara.vega@frb.gov. We are grateful to EBS/ICAP for providing the data, and to Nicholas Klagge and James S. Hebden for their excellent research assistance. We would like to thank Cam Harvey, an anonymous Associate Editor, an anonymous Advisory Editor, and an anonymous referee for their valuable comments. We also benefited from the comments of Gordon Bodnar, Charles Jones, Terrence Hendershott, Lennart Hjalmarsson, Luis Marques, Albert Menkveld, Dagfinn Rime, Alec Schmidt, John Schoen, Noah Stoffman, and of participants in the University of Washington Finance Seminar, SEC Finance Seminar Series, Spring 2009 Market Microstructure NBER conference, San Francisco AEA 2009 meetings, the SAIS International Economics Seminar, the SITE 2009 conference at Stanford, the Barcelona EEA 2009 meetings, the Essex Business School Finance Seminar, the EDHEC Business School Finance Seminar, the Imperial College Business School Finance Seminar, and the 2013 Paris High Frequency Trading conference. The views in this paper are solely the responsibility of the authors and should not be interpreted as reflecting the views of the Board of Governors of the Federal Reserve System or of any other person associated with the Federal Reserve System.

1 Introduction

The use of algorithmic trading (AT), where computers monitor markets and manage the trading process at high frequency, has become common in major financial markets in recent years, beginning in the U.S. equity market in the 1990s. Since the introduction of algorithmic trading, there has been widespread interest in understanding the potential impact it may have on market dynamics, particularly recently following several trading disturbances in the equity market blamed on computer-driven trading. While some have highlighted the potential for more efficient price discovery, others have expressed concern that it may lead to higher adverse selection costs and excess volatility.¹ In this paper, we analyze the effect algorithmic (“computer”) trades and non-algorithmic (“human”) trades have on the informational efficiency of foreign exchange prices; it is the first formal empirical study on the subject in the foreign exchange market.

We rely on a novel data set consisting of several years (September 2003 to December 2007) of minute-by-minute trading data from Electronic Broking Services (EBS) in three currency pairs: the euro-dollar, dollar-yen, and euro-yen. The data represent a large share of spot interdealer transactions across the globe in these exchange rates, with EBS widely considered to be the primary site of price discovery in these currency pairs during our sample period. A crucial feature of the data is that, on a minute-by-minute frequency, the volume and direction of human and computer trades are explicitly identified, allowing us to measure their respective impacts at high frequency. Another useful feature of the data is that it spans the introduction and rapid growth of algorithmic trading in an important market where it had not been previously allowed.

The theoretical literature highlights two main differences between computer and human traders. First, computers are faster than humans, both in processing information and in acting on that information. Second, there is the potential for higher correlation in computers’ trading actions than in those of humans, as computers need to be pre-programmed and may react similarly to a given signal. There is no agreement, however, on the impact that these features of algorithmic trading may have on the price discovery process.

Biais, Foucault, and Moinas (2011) and Martinez and Rosu (2011) argue that the speed advantage of algorithmic traders over humans – specifically their ability to react more quickly to public information – should have a positive effect on the informativeness of prices. In their theoretical models, algorithmic traders are better informed than humans and use market orders to exploit their information. Given these assumptions, the authors show that the presence of algorithmic traders makes asset prices more informationally efficient, but, importantly, that their trades are a source of adverse selection for those who provide liquidity. They argue that algorithmic traders contribute to price discovery because, once price inefficiencies arise, AT quickly makes them disappear by trading on posted quotes. Similarly Oehmke (2009) and Kondor (2009), in the

¹Biais and Woolley (2011) and Foucault (2012) provide excellent surveys of the recent literature on this topic.

context of traders who implement “convergence trade” strategies (i.e., strategies taking long-short positions in assets with identical underlying cash flows that temporarily trade at different prices), argue that the higher the number of traders who implement those strategies, the more efficient prices will be. One could also advance, as does Hoffmann (2012), that better informed algorithmic traders who specialize in providing liquidity make prices more informationally efficient by posting quotes that reflect new information quickly, thus preventing arbitrage opportunities from occurring in the first place.

In contrast to these mostly positive views on algorithmic trading and price efficiency, Foucault, Hombert, and Rosu (2012) argue that, in a world with no asymmetric information, the speed advantage of algorithmic traders would not increase the informativeness of prices but would still increase adverse selection costs. Biais and Woolley (2011), in their survey article, point out that the potential commonality of trading actions amongst computers may have a negative effect on the informativeness of prices. Khandani and Lo (2007, 2011), who analyze the large losses that occurred for many quantitative long-short equity strategies at the beginning of August 2007, highlight the possible adverse effects on the market of such commonality in behavior across market participants (algorithmic or not) and provide empirical support for this concern.² Kozhan and Wah Tham (2012), in contrast to Oehmke’s (2009) and Kondor’s (2009) more standard notion that competition improves price efficiency, argue that computers entering the same trade at the same time to exploit an arbitrage opportunity could cause a crowding effect, which in turn would push prices away from their fundamental values. Stein (2009) also highlights this crowding effect in the context of hedge funds simultaneously implementing “convergence trade” strategies.

Guided by this literature, our paper studies the impact of algorithmic trading on the price discovery process in the foreign exchange market. Besides giving us a measure of the relative participation of computers and humans in trades, our data also allows us to study more precisely how algorithmic trading affects price discovery. We study separately, in particular, the impact of trades where computers are providing liquidity to the market and trades where computers are taking liquidity from the market, addressing one of the questions posed in the literature. We also look at how the share of the market’s order flow generated by algorithmic traders impacts price discovery. Finally, to address another concern highlighted in the literature—namely that the trading strategies used by computers are more correlated than those used by humans, potentially creating excessive volatility—we propose a novel way of indirectly inferring the correlation among computer trading strategies from our trading data. The primary idea behind the measure that we design is that traders who follow similar trading strategies will trade less with each other than those who follow less correlated strategies.

²One could also argue that if algorithmic traders behave, as a group, like the positive-feedback traders of DeLong, Shleifer, Summers, and Waldman (1990) or the chartists described in Froot, Scharfstein, and Stein (1992), or the short-term investors in Vives (1995), then their activity, in total, would likely create excess volatility.

We first use our data to study the impact of algorithmic trading on the frequency of triangular arbitrage opportunities amongst the euro-dollar, dollar-yen, and euro-yen currency pairs. These arbitrage opportunities are a clear example of prices not being informationally efficient in the foreign exchange market. We document that the introduction and growth of algorithmic trading coincided with a substantial reduction in triangular arbitrage opportunities. We then continue with a formal analysis of whether algorithmic trading activity causes a reduction in triangular arbitrage opportunities, or whether the relationship is merely coincidental, possibly due to a concurrent increase in trading volume or decrease in price volatility. To that end, we formulate a high-frequency vector autoregression (VAR) specification that models the interaction between triangular arbitrage opportunities and algorithmic trading (the overall share of algorithmic trading and the various facets of algorithmic activity discussed above). In the VAR specification, we control for time trends, trading volume, and exchange rate return volatility in each currency pair. We estimate both a reduced form of the VAR and a structural VAR which uses the heteroskedasticity identification approach developed by Rigobon (2003) and Rigobon and Sack (2003, 2004).³ In contrast to the reduced-form Granger causality tests, which essentially measure predictive relationships, the structural VAR estimation allows for an identification of the contemporaneous causal impact of algorithmic trading on triangular arbitrage opportunities.

Both the reduced-form and structural-form VAR estimations show that algorithmic trading activity causes a reduction in the number of triangular arbitrage opportunities. In addition, we find that algorithmic traders reduce arbitrage opportunities more by acting on the quotes posted by non-algorithmic traders than by posting quotes that are then traded upon. This result is consistent with the view that algorithmic trading improves informational efficiency by speeding up price discovery, but that, at the same time, it increases the adverse selection costs to slower traders, as suggested by the theoretical models of Biais, Foucault, and Moinas (2011) and Martinez and Rosu (2011). We also find that a higher degree of correlation amongst algorithmic trading strategies reduces the number of arbitrage opportunities over our sample period. Thus, contrary to Kozhan and Wah Tham (2012), in this particular example commonality in trading strategies appears to be beneficial to the efficiency of the price discovery process.

The impact of algorithmic trading on the frequency of triangular arbitrage opportunities is, however, only one facet of how computers may affect the price discovery process. More generally, we investigate whether algorithmic trading contributes to the temporary deviation of asset prices from their fundamental values, resulting in excess volatility, particularly at high frequencies. To investigate the effect of algorithmic trading on this aspect of informational efficiency, we study the autocorrelation of high-frequency returns in our three exchange rates.⁴ We again estimate both a reduced form and a structural VAR and find that, on average, an

³To identify the parameters of the structural VAR, we use the heteroskedasticity in algorithmic trading activity across the sample.

⁴We use the term “excess volatility” to refer to any variation in prices that is not solely reflecting changes in the underlying

increase in algorithmic trading participation in the market causes a reduction in the degree of autocorrelation of high-frequency returns. Interestingly, we find that the improvement in the informational efficiency of prices now seems to come predominantly from an increase in the trading activity of algorithmic traders when they are providing liquidity—that is, posting quotes which are hit—not from an increase in the trading activity of algorithmic traders who hit posted quotes. In other words, in this case, in contrast to the study of triangular arbitrage, algorithmic traders appear to increase the informational efficiency of prices by posting quotes that reflect new information more quickly, consistent with Hoffmann (2012).

Also in contrast to the triangular arbitrage case, we do not find a statistically significant association between a higher correlation of algorithmic traders' actions and high-frequency return autocorrelation. The difference between the autocorrelation results and those for triangular arbitrage may offer some support for one of the conclusions of Foucault (2012): The effect of algorithmic trading on the informativeness of prices may ultimately depend on the type of strategies used by algorithmic traders, not on the presence of algorithmic trading per se.

The paper proceeds as follows. In Section 2, we briefly discuss the related empirical literature on the effect of algorithmic trading on market quality. Section 3 introduces the high-frequency data used in this study, including a short description of the structure of the market and an overview of the growth of algorithmic trading in the foreign exchange market over time. In Section 4, we present our measures of the various aspects of algorithmic trading, including whether actions by computer traders appear to be more correlated than those of human traders. In Sections 5 and 6, we test whether there is evidence that algorithmic trading activity has a causal impact on the informativeness of prices, looking first at triangular arbitrage and then at high-frequency return autocorrelation. Finally, Section 7 concludes. Some additional clarifying and technical material is found in the Appendix.

2 Related Empirical Literature

Our study contributes to the new empirical literature on the impact of algorithmic trading on various measures of market quality, with the vast majority of the studies focusing on equity markets. A number of these studies use proxies to measure the share of algorithmic trading in the market. Hendershott, Jones, and Menkveld (2011), in one of the earliest studies, use, for instance, the flow of electronic messages on the NYSE after the implementation of Autoquote as a proxy for algorithmic trading. They find that algorithmic trading improves standard measures of liquidity (the quoted and effective bid-ask spreads). They attribute the decline

efficient price process. Under the usual assumption that efficient prices follow a random walk specification (see discussion in Footnote 23), autocorrelation in returns therefore imply some form of excess volatility and temporary deviations from the efficient price. When discussing the empirical results, we use the terms autocorrelation and excess volatility interchangeably.

in spreads to a decline in adverse selection — a decrease in the amount of price discovery associated with trading activity and an increase in the amount of price discovery that occurs without trading. The authors’ interpretation of the empirical evidence is that computers enhance the informativeness of quotes by more quickly resetting their quotes after news arrivals. Boehmer, Fong, and Wu (2012) use a similar identification strategy to that of Hendershott, Jones, and Menkveld (2011). Rather than using the implementation of Autoquote, however, they use the first availability of co-location facilities around the world to identify the effect algorithmic trading activity has on liquidity, short-term volatility, and the informational efficiency of stock prices. Hendershott and Riordan (2012) use one month of algorithmic trading data in the 30 DAX stocks traded on the Deutsche Boerse and find that algorithmic traders improve market liquidity by providing liquidity when it is scarce, and consuming it when it is plentiful.

Some recent studies have focused more specifically on the impact of high-frequency trading (HFT), generally viewed as a subset of algorithmic trading (not all algorithmic traders trade at extremely high frequency). For instance Brogaard, Hendershott, and Riordan (2012) use NASDAQ data from 2008 and 2009 and find that high-frequency traders (HFTs) facilitate price efficiency by trading in the direction of permanent price changes during macroeconomic news announcement times and at other times. Hirschey (2011), using a similar database to that of Brogaard, Hendershott, and Riordan (2012), finds that HFT’s aggressive purchases (sales) predict future aggressive purchases (sales) by non-HFTs. Both of these studies suggest that HFTs are the “informed” traders. Hasbrouck and Saar (2012) develop an algorithm to proxy for overall HFT activity and conclude that HFT activity reduces short-term volatility and bid-ask spreads, and that it increases displayed depth in the limit order book. Menkveld (2011) analyzes the trading strategy of one large HFT in the Chi-X market and concludes that the HFT behaves like a fast version of the classic market maker. Kirilenko, Kyle, Samadi, and Tuzun (2011) study the role HFTs played during the May 6, 2010 “flash crash” and conclude that HFTs did not trigger the crash, but that they contributed to the crash by pulling out of the market as market conditions became challenging.

Our work complements these studies in several dimensions. We study, for the first time, a different asset class, foreign exchange, which is traded in a very large global market.⁵ We have a long data set which clearly identifies computer-generated trades without appealing to proxies, and which spans the introduction of algorithmic trading in the market. The data also permits us to differentiate the effects of certain features

⁵The structure of the wholesale spot foreign exchange market is obviously different from that of the major equity markets, but some of the differences probably facilitate the analysis of the impact of AT. In particular, while there are several platforms where one can trade foreign exchange in large quantities, price discovery in the currency pairs we study occurs overwhelmingly on one platform, EBS, the source of our data. In addition, unlike a number of equity trading venues, traders do not get rebates for providing liquidity on EBS, which may allow for a “cleaner” study of the role of AT making and taking in price discovery. Finally, in the foreign exchange market, AT potentially causes a smaller increase in adverse selection costs than it may cause in the equity market because it is a wholesale market. The minimum trade size is one million of the base currency, and thus there are no “small” market participants.

of algorithmic trading, separating, for instance, trades initiated by a computer and trades initiated by a human, which in turn allows us to address recently-developed theories on how algorithmic trading impacts price discovery. We have data on three interrelated exchange rates, and therefore can analyze the impact of computer trades on the frequency of a very obvious type of arbitrage opportunity. And, finally, we study the correlation of algorithmic trading strategies and its impact on the informational efficiency of prices, also relating our findings to the recent theoretical literature.

3 Data Description

3.1 Market Structure

Over our sample period, from 2003 to 2007, two electronic platforms process a majority of global interdealer spot trading in the major currency pairs, one offered by Reuters, and one offered by Electronic Broking Services (EBS). Both of these trading platforms are electronic limit order books. Importantly, trading in each major currency pair is highly concentrated on only one of the two systems. Of the most traded currency pairs (exchange rates), the top two, euro-dollar and dollar-yen, trade primarily on EBS, while the third, sterling-dollar, trades primarily on Reuters. As a result, price discovery for spot euro-dollar, for instance, occurs on the EBS system, and dealers across the globe base their spot and derivative quotes on that price. EBS controls the network and each of the terminals on which the trading is conducted. Traders can enter trading instructions manually, using an EBS keyboard, or, upon approval, via a computer directly interfacing with the system. The type of trader (human or computer) behind each trading instruction is recorded by EBS, allowing for our study.

The EBS system is an interdealer system accessible to foreign exchange dealing banks and, under the auspices of dealing banks (via prime brokerage arrangements), to hedge funds and commodity trading advisors (CTAs). As it is a “wholesale” trading system, the minimum trade size over our sample period is 1 million of the “base” currency, and trade sizes are only allowed in multiple of millions of the base currency. We analyze data in the three most-traded currency pairs on EBS, euro-dollar, dollar-yen, and euro-yen.⁶

3.2 Quote and Transaction Data

Our data consists of both quote data and transactions data. The quote data, at the one-second frequency, consist of the highest bid quote and the lowest ask quote on the EBS system in our three currency pairs. The quote data are available from 1997 through 2007. All the quotes are executable and therefore truly represent

⁶The euro-dollar currency pair is quoted as an exchange rate in dollars per euro, with the euro the “base” currency. Similarly, the euro is also the base currency for euro-yen, while the dollar is the base currency for the dollar-yen pair.

the market price at that instant. From these data, we construct mid-quote series from which we compute exchange rate returns at various frequencies. The transactions data, available from October 2003 through December 2007, are aggregated by EBS at the one-minute frequency. Throughout the subsequent analysis, we focus on data sampled between 3am and 11am New York time, which represent the most active trading hours of the day (see Berger et al., 2008, for further discussion on trading activity on the EBS system). That is, each day in our sample is made up of the intra-daily observations between 3am and 11am New York time.⁷

The transaction data provide detailed information on the volume and direction of trades that can be attributed to computers and humans in each currency pair. Specifically, each minute we observe trading volume and order flow for each of the four possible pairs of human and computer makers and takers: human-maker/human-taker (HH), computer-maker/human-taker (CH), human-maker/computer-taker (HC), and computer-maker/computer-taker (CC).⁸ Order flow is defined, as is common, as the net of buyer-initiated trading volume minus seller-initiated trading volume, with traders buying or selling the base currency. We denote the trading volume and order flow attributable to any maker-taker pair as $Vol(\cdot)$ and $OF(\cdot)$, respectively.

Figure 1 shows, from 2003 through 2007, for each currency pair, the percent of trading volume where at least one of the two counterparties is an algorithmic trader. We label this variable $VAT = 100 \times \frac{Vol(CH+HC+CC)}{Vol(HH+CH+HC+CC)}$. From its beginning in late 2003, the fraction of trading volume involving algorithmic trading for at least one of the counterparties grows by the end of 2007 to near 60 percent for euro-dollar and dollar-yen trading, and to about 80 percent for euro-yen trading.

Figure 2 shows the evolution over time of the four different possible types of trades: $Vol(HH)$, $Vol(CH)$, $Vol(HC)$, and $Vol(CC)$, as fractions of the total volume. By the end of 2007, in the euro-dollar and dollar-yen markets, human to human trades, the solid lines, account for slightly less than half of the volume, and computer to computer trades, the dotted lines, for about ten to fifteen percent. In these two currency pairs, $Vol(CH)$ is often close to $Vol(HC)$, i.e., computers take prices posted by humans, the dashed lines, about as often as humans take prices posted by market-making computers, the dotted-dashed lines. The story is different for the cross-rate, the euro-yen currency pair. By the end of 2007, there are more computer to computer trades than human to human trades. But the most common type of trade in euro-yen is computers trading on prices posted by humans. We believe this reflects computers taking advantage of short-lived

⁷We exclude data collected from Friday 17:00 through Sunday 17:00 New York time from our sample, as activity on the system during these “non-standard” hours is minimal and not encouraged by the foreign exchange community. Trading is continuous outside of the weekend, but the value date for trades, by convention, changes at 17:00 New York time, which therefore marks the end of each trading day. We also drop certain holidays and days of unusually light volume: December 24-December 26, December 31-January 2, Good Friday, Easter Monday, Memorial Day, Labor Day, Thanksgiving and the following day, and July 4 (or, if this is on a weekend, the day on which the U.S. Independence Day holiday is observed).

⁸The naming convention for “maker” and “taker” reflects the fact that the “maker” posts quotes before the “taker” chooses to trade at that price. Posting quotes is, of course, the traditional role of the market-“maker.” The taker is also often viewed as the “aggressor” or the “initiator” of the trade. We refer the reader to Appendix A1 for more details on how we calculate volume and order flow for these four possible pairs of human and computer makers and takers.

triangular arbitrage opportunities, where prices set in the euro-dollar and dollar-yen markets, the primary sites of price discovery, are very briefly out of line with the euro-yen cross rate. Detecting and trading on triangular arbitrage opportunities is widely thought to have been one of the first strategies implemented by algorithmic traders in the foreign exchange market, which is consistent with the more rapid growth in algorithmic activity in the euro-yen market documented in Figure 1. We discuss the evolution of the frequency of triangular arbitrage opportunities in Section 5 of the paper.

4 Measuring Various Features of Algorithmic Trading

4.1 Inferring The Correlation of Algorithmic Strategies from Trade Data: The *R*-Measure

There is little solid information, and certainly no data, about the precise mix of strategies used by algorithmic traders in the foreign exchange market, as traders and EBS keep what they know confidential. From conversations with market participants, however, we believe that, over our sample period, about half of the algorithmic trading volume on EBS comes from traditional foreign exchange dealing banks, with the other half coming from hedge funds and commodity trading advisors (CTAs). Hedge funds and CTAs, who access EBS under prime-brokerage arrangements, can only trade algorithmically over our sample period. Some of the banks' computer trading in our sample is related to activity on their own customer-to-dealer platforms, to automated hedging activity, and to the optimal execution of large orders. But a sizable fraction (perhaps almost a half) is believed to be proprietary trading using a mix of strategies similar to what hedge funds and CTAs use. These strategies include various types of high-frequency arbitrage, including some across different asset markets, a number of lower-frequency statistical arbitrage strategies (including carry trades), and strategies designed to automatically react to news and data releases (still fairly rare in the foreign exchange market by 2007). Overall, market participants believe that the main difference between the mix of algorithmic strategies used in the foreign exchange market and the mix used in the equity market is that optimal execution algorithms are less prevalent in foreign exchange than in equity.

As mentioned previously, a concern expressed by the literature about algorithmic traders is that their strategies, which have to be pre-programmed, may be less diverse, more correlated, than those of humans.⁹ Computers could then react in the same fashion, at the same time, to the same information, creating excess volatility in the market, particularly at high frequency. While we do not observe the trading strategies of our market participants, we can derive some information about the correlation of algorithmic strategies from the

⁹As pointed out by the referee, it is true that algorithms are necessarily rule-based at very high execution frequencies, but at a somewhat lower frequency, human monitoring is still common for a variety of algorithms.

trading activity of computers and humans. The intuitive idea is the following. Traders who follow similar trading strategies and therefore send similar trading instructions at the same time, will trade less with each other than those who follow less correlated strategies. Therefore, the extent to which computers trade (or do not trade) with each other should contain information about how correlated their algorithmic strategies are.

More precisely, to extract this information, we first consider a simple benchmark model that assumes random and independent matching of traders. This is a reasonable assumption given the lack of discrimination between keyboard traders and algorithmic traders in the EBS matching process; that is, EBS does not differentiate in any way between humans and computers when matching buy and sell orders in its electronic order book. Traders also do not know the identity and type of trader they have been matched with until after the full trade is complete. The model allows us to determine the theoretical probabilities of the four possible types of trades: Human-maker/human-taker, computer-maker/human-taker, human-maker/computer-taker and computer-maker/computer-taker. We then compare these theoretical probabilities to those observed in the actual trading data. The benchmark model is fully described in Appendix A2, and below we outline the main concepts and empirical results.

Under our random and independent matching assumption, computers and humans, both of which are indifferent *ex-ante* between making and taking, trade with each other in proportion to their relative presence in the market. For instance, in a world with more human trading activity than computer trading activity (which is the case in the vast majority of our sample), we should observe that computers take more liquidity from humans than from other computers. That is, the probability of observing human-maker/computer-taker trades, $Prob(HC)$, should be larger than the probability of observing computer-maker/computer-taker trades, $Prob(CC)$. We label the ratio of the two, $Prob(HC)/Prob(CC)$, the computer-taker ratio, RC . Similarly, under our assumption, in such a world one would expect humans to take more liquidity from other humans than from computers, i.e., $Prob(HH)$ should be larger than $Prob(CH)$. We label this ratio, $Prob(HH)/Prob(CH)$, the human-taker ratio, RH . In summary, in our data with more human trading activity than computer trading activity, one would thus expect to find that $RC > 1$ and $RH > 1$.

Importantly however, irrespective of the relative prevalence of human and computer trading activity, the benchmark model predicts that the ratio of these two ratios, the computer-taker ratio divided by the human-taker ratio, should be equal to one. That is, under random matching, the model predicts $R = RC/RH = 1$, with computers taking liquidity from humans in the same *proportion* that humans take liquidity from other humans.

The R -measure described above implicitly takes into account the commonality of trading direction to the extent that the matching process of the electronic limit order book takes it into account. For instance, two computers sending at the same time instructions to buy the euro will obviously not be matched by the

electronic order book. In contrast, if about half of the algorithmic traders are buying and about half are selling, algorithmic traders will have a high probability of being matched with each other. Still, the buying and selling behavior of computers is an additional variable our data allows us to condition on when we infer whether computers' trading actions are highly correlated.¹⁰ Hence, we also describe in detail in Appendix A2 a model that explicitly takes the trading direction of computers into account. Using notation similar to the model without trading direction, this model yields the ratio of the buy ratios, $R^B = RC^B / RH^B$, and the ratio of the sell ratios, $R^S = RC^S / RH^S$. Again, under random matching, the benchmark model predicts that these ratios will both be equal to one.

Based on the model described above, we calculate R , R^S , and R^B for the three currency pairs in our 2003-2007 data at three different frequencies: 1-minute, 5-minute, and daily. Specifically, the realized values of RH and RC are given by $\widehat{RH} = \frac{Vol(HH)}{Vol(CH)}$ and $\widehat{RC} = \frac{Vol(HC)}{Vol(CC)}$, where, for instance, $Vol(HC)$ is either the 1-min, 5-min or daily trading volume between human makers and computer takers, following the notation described in Section 3. Similarly, we define $\widehat{RH^S} = \frac{Vol(HH^S)}{Vol(CH^S)}$, $\widehat{RH^B} = \frac{Vol(HH^B)}{Vol(CH^B)}$, $\widehat{RC^S} = \frac{Vol(HC^S)}{Vol(CC^S)}$, and $\widehat{RC^B} = \frac{Vol(HC^B)}{Vol(CC^B)}$, where $Vol(HH^B)$ is the buy volume between human makers and human takers (i.e., the trading volume which involves buying of the base currency by the taker), $Vol(HH^S)$ is the sell volume between human makers and human takers, and so forth.

Table 1 shows the means of the natural log of the 1-min, 5-min, and daily ratios of ratios, $\ln \widehat{R} = \ln(\frac{\widehat{RC}}{\widehat{RH}})$, $\ln \widehat{R^S} = \ln(\frac{\widehat{RC^S}}{\widehat{RH^S}})$, and $\ln \widehat{R^B} = \ln(\frac{\widehat{RC^B}}{\widehat{RH^B}})$, calculated for each currency pair in our data.¹¹ In contrast to the benchmark predictions that $R \equiv 1$, $R^B \equiv 1$ and $R^S \equiv 1$, or equivalently that $\ln R \equiv 0$, $\ln R^B \equiv 0$ and $\ln R^S \equiv 0$, we find that, for all three currency pairs, at all frequencies, $\ln \widehat{R}$, $\ln \widehat{R^B}$ and $\ln \widehat{R^S}$ are substantially and significantly greater than zero. The table also shows the number of periods in which the statistics are above zero. In all currencies, at the daily frequency, $\ln \widehat{R}$, $\ln \widehat{R^B}$ and $\ln \widehat{R^S}$ are above zero for more than 95 percent of the days. At all frequencies, the readings are highest for the euro-yen currency pair.¹²

There is clear evidence, therefore, that the pattern of trading among humans and computers is different from that predicted by a random matching model. In particular, our finding that R is greater than one indicates that computers do not trade with each other as much as a random matching model would predict. This

¹⁰We thank an anonymous referee for this suggestion.

¹¹We report summary statistics for the log-transformed variables because these results are invariant to whether one defines the R -measures as above, with a value greater than one consistent with correlated computer trading, or in the opposite direction (having $Vol(CC)$ in the numerator), with a value less than one consistent with correlated computer trading. That is, the mean estimate will not be symmetric around one when inverting the R -measure, whereas the mean estimate of $\ln(R)$ will be symmetric around zero when inverting the R -measure.

¹²Note that the fraction of missing observation increases in all currency pairs as the period of observation over which the R -measure is calculated gets smaller. Missing observations arise mostly when $Vol(CC) = 0$, that is when computers do not trade with each other at all in a given time interval, which is more likely to occur in a shorter time interval. This can happen because computer trading activity is very small, as it is early in the sample, reducing the probability of a CC match. But it can also happen when there is more abundant computer trading activity but their strategies are so highly correlated that no CC match occurs. Excluding the information contained in this second type of missing observations means that, if anything, we may be underestimating the extent of the correlation of algorithmic strategies in the data.

finding is consistent with the trading strategies of computers being correlated, and this is the interpretation that we propose. However, we note that a finding of $R > 1$ is not unambiguous proof that computer strategies are correlated. Indeed, as a consequence of market clearing, $R > 1$ must also imply that humans do not trade with each other as much as the random matching model predicts, which could also reasonably be viewed as consistent with correlated human strategies. In each case, there could be an endogenous response from one group (humans or computers) to the correlated strategies of the other group, such that trading within each group is diminished. The fundamental cause that forces R to deviate from one is therefore not identified.

Still, we believe that the interpretation of $R > 1$ as evidence of correlated AT strategies is the most plausible. From conversations with market participants, there is widespread anecdotal evidence that in the very first years of AT in this market, a fairly limited number of strategies was being implemented, with triangular arbitrage among the most prominent. As mentioned previously, Table 1 shows that the estimates of R are always highest for the euro-yen currency pair. This is consistent with the view that a large fraction of computer traders in the cross-rate were often attempting to take advantage of the same short-lived triangular arbitrage opportunities, and thus were more likely to send the same trading instructions at the same time in this currency pair, trading little with each other.¹³ Furthermore, human participation remains higher than computer participation in almost all of our sample and it seems less plausible that a large number of market participants are using the same strategies.¹⁴

We next describe the other variables that we create to measure certain features of algorithmic trading activity and that we will also use in our analysis.

4.2 Measuring Other Features of Algorithmic Trading Activity

With our trading data providing information, minute by minute, on the volume of trade, direction of trade, and initiating of the trade attributed to algorithmic (computer) and non-algorithmic (human) traders, we can study not just the overall impact of the presence of algorithmic trading in the market but also the mechanism by which algorithmic trading affects the market. To this effect, we create several variables at a minute-by-minute frequency to use in the analysis. (1) We measure overall algorithmic trading activity as the fraction of total volume where a computer is at least one of the two counterparts of a trade. As explained before, we label this variable $VAT = 100 \times \frac{Vol(HC)+Vol(CH)+Vol(CC)}{Vol(HH)+Vol(HC)+Vol(CH)+Vol(CC)}$ and show its evolution over time in Figure 1. This is a more precise replacement for the various proxies for algorithmic trading activity that have been used in some recent studies, and equivalent to what has been used in the studies which had access to

¹³Euro-yen is the currency pair with the smallest trading volume in our sample. A given triangular arbitrage trade, which automatically involves a similarly sized trade in our three exchange rates, would therefore represent a larger fraction of the total trades in euro-yen.

¹⁴Human participation remains higher than computer participation throughout our sample for euro-dollar and dollar-yen. In euro-yen, the cross rate, only the last few months of the 2003-2007 sample have a higher computer participation.

actual data on the fraction of algorithmic trading in the market. (2) We measure the fraction of the overall trading volume where a computer is the aggressor in a trade, trading on an existing quote, in other words the relative taking activity of computers. We label this variable $V Ct = 100 \times \frac{Vol(HC) + Vol(CC)}{Vol(HH) + Vol(HC) + Vol(CH) + Vol(CC)}$. (3) We measure the fraction of the overall trading volume where a computer provides the quote hit by another trader, in other words the relative making activity of computers. We label this variable $V Cm = 100 \times \frac{Vol(CH) + Vol(CC)}{Vol(HH) + Vol(HC) + Vol(CH) + Vol(CC)}$. (4) We measure the share of order flow in the market coming from computer traders, relative to the order flow of both computer and human traders. This variable, which, by the definition of order flow, takes into account the sign of the trades initiated by computers, can be viewed as a measure of the relative “directional intensity” of computer-initiated trades in the market. We label this variable $OF Ct = 100 \times \frac{|OF(C-Take)|}{|OF(C-Take)| + |OF(H-Take)|}$, where $OF(C-Take) = OF(HC) + OF(CC)$ and $OF(H-Take) = OF(HH) + OF(CH)$.¹⁵

We will use these variables in addition to our R -measure to study more precisely how AT affects the process of price discovery. For instance, in our analysis of triangular arbitrage reported below, a finding that $V Ct$ is more strongly (negatively) related to triangular arbitrage opportunities than $V Cm$ would lead us to conclude that the empirical evidence provides some support for the conclusions of Biais, Foucault and Moinas (2011), in that algorithmic traders contribute to market efficiency by trading on existing arbitrage opportunities. An opposite finding would be evidence that it is instead the algorithmic traders who specialize in providing liquidity that make prices more informationally efficient, by posting quotes that reflect information more quickly in the first place.

5 Triangular Arbitrage and AT

In this section we analyze the effects of algorithmic trading on the frequency of triangular arbitrage opportunities in the foreign exchange market. We begin by providing some suggestive graphical evidence that the introduction and growth of AT coincides with a reduction in triangular arbitrage opportunities, and then proceed with a more formal causal analysis.

5.1 Preliminary Graphical Evidence

Our data contain second-by-second bid and ask quotes on three related exchange rates (euro-dollar, dollar-yen, and euro-yen), allowing us to estimate the frequency with which these exchange rates are “out of alignment.”

¹⁵ One could also consider an analogously defined $OF Cm$ variable as a measure of how correlated the actions of liquidity-making computers are. However, we chose not to add this variable to our analysis because the interpretation of the results seems ambiguous. For instance, if one found that an increase in $OF Cm$ led to an increase in high-frequency return autocorrelation, we would still be hesitant to attribute this finding to the correlated actions of liquidity-making computers. Instead, because of the empirical evidence which shows that order flow from the liquidity taker’s point of view drives asset prices, it would be more likely that that finding is driven by the traders who are taking liquidity from computers in a correlated way.

More precisely, each second we evaluate whether a trader, starting with a dollar position, could profit from purchasing euros with dollars, purchasing yen with euros, and purchasing dollars with yen, all simultaneously at the relevant bid and ask prices. An arbitrage opportunity is recorded for any instance when such a strategy (or a “round trip” in the other direction) yields a strictly positive profit. We show in Figure 3 (top panel) the daily frequency of such triangular arbitrage opportunities from 2003 through 2007 as well as the frequency of triangular arbitrage opportunities that yield a profit of 1 basis point or more (bottom panel).¹⁶

The frequency of arbitrage opportunities clearly drops over our sample. For arbitrage opportunities yielding a profit of one basis point or more (Figure 3 bottom panel), the drop is particularly noticeable around 2005, when the rate of growth in algorithmic trading is highest. In fact, by 2005 we begin to see entire days without arbitrage opportunities of that magnitude. On average in 2003 and 2004, the frequency of such arbitrage opportunities is about 0.5 percent. By 2007, at the end of our sample, the frequency has declined to 0.03 percent. As seen in the top panel of Figure 3, the decline in the frequency of arbitrage opportunities with a profit strictly greater than zero is, as we would expect, more gradual, and the frequency does not approach zero by the end of our sample. In 2003 and 2004, these opportunities occur with a frequency of about 3 percent. By 2007, their frequency has dropped to about 1.5 percent. This simple analysis highlights the potentially important impact of algorithmic trading in this market, but obviously does not prove that algorithmic trading caused the decline, as other factors could have contributed to, or even driven, the drop in arbitrage opportunities. However, the findings line up well with (i) the anecdotal (but widespread) evidence that one of the first strategies implemented by algorithmic traders in the foreign exchange market aimed to detect and profit from triangular arbitrage opportunities and with (ii) the view that, as more arbitrageurs attempt to take advantage of these opportunities, their observed frequency declines (Oehmke (2009) and Kondor (2009)).

5.2 A Formal Analysis

We now attempt to more formally identify the relationship between algorithmic trading and the frequency of triangular arbitrage opportunities, controlling for changes in the total volume of trade and market volatility, as well as general time trends in the data. We use the second-by-second quote data to construct a minute-by-minute measure of the frequency of triangular arbitrage opportunities. Specifically, following the approach described in the preceding section, for each second we calculate the maximum profit achievable from a “round trip” triangular arbitrage trade in either direction, starting with a dollar position. A minute-by-

¹⁶We take into account actual bid and ask prices in our triangular arbitrage calculations every second, which implies that the largest component of the transactions costs is accounted for in the calculations. But algorithmic traders likely incur other, more modest, costs, some fixed, some variable, but with the additional variable costs certainly well below 1 basis point in this market. We should not, however, expect all arbitrage opportunities with a positive profit to completely disappear. In the formal analysis we use a straightforward zero minimum profit level to calculate the frequency of arbitrage opportunities.

minute measure of the frequency of triangular arbitrage opportunities is then calculated as the number of seconds within each minute with a strictly positive profit.

Algorithmic trading and triangular arbitrage opportunities are likely determined simultaneously, in the sense that both variables have a contemporaneous impact on each other. OLS regressions of contemporaneous triangular arbitrage opportunities on contemporaneous algorithmic trading activity are therefore likely biased and misleading. In order to overcome these potential difficulties, we estimate a VAR system that allows for both Granger causality tests as well as contemporaneous identification through heteroskedasticity in the data across the sample period (Rigobon (2003) and Rigobon and Sack (2003, 2004)). Let Arb_t be the minute-by-minute measure of triangular arbitrage possibilities and AT_t^{avg} be average AT activity across the three currency pairs; AT_t^{avg} is used to represent either of the five previously defined measures of AT activity. Define $Y_t = (Arb_t, AT_t^{avg})$ and the structural VAR system is

$$AY_t = \Phi(L)Y_t + \Lambda X_{t-1:t-20} + \Psi G_t + \epsilon_t. \quad (1)$$

Here A is a 2×2 matrix specifying the contemporaneous effects, normalized such that all diagonal elements are equal to 1. $\Phi(L)$ is a lag-function that controls for the effects of the lagged endogenous variables. $X_{t-1:t-20}$ consists of lagged control variables not modelled in the VAR. Specifically, $X_{t-1:t-20}$ includes the sum of the volume of trade in each currency pair over the past 20 minutes and the volatility in each currency pair over the past 20 minutes, calculated as the sum of absolute returns over these 20 minutes. G_t represent a set of deterministic functions of time t , capturing individual trends and intra-daily patterns in the variables in Y_t . In particular, $G_t = (I_{\{t \in 1st \text{ month of sample}\}}, \dots, I_{\{t \in \text{last month of sample}\}}, I_{\{t \in 1st \text{ half-hour of day}\}}, \dots, I_{\{t \in \text{last half-hour of day}\}})$, captures long term secular trends in the data by year-month dummy variables, as well as intra-daily patterns accounted for by half-hour dummy variables. The structural shocks to the system are given by ϵ_t , which, in line with standard structural VAR assumptions, are assumed to be independent of each other and serially uncorrelated at all leads and lags. The number of lags included in the VAR is set to 20. Equation (1) thus provides a very general specification, allowing for full contemporaneous interactions between triangular arbitrage opportunities and AT activity.

Before describing the estimation of the structural system, we begin the analysis with the reduced form system,

$$Y_t = A^{-1}\Phi(L)Y_t + A^{-1}\Lambda X_{t-1:t-20} + A^{-1}\Psi G_t + A^{-1}\epsilon_t. \quad (2)$$

The reduced form is estimated equation-by-equation using ordinary least squares, and Granger causality tests are performed to assess the effect of AT on triangular arbitrage opportunities (and vice versa). In particular,

we test whether the *sum* of the coefficients on the lags of the causing variable is equal to zero. Since the sum of the coefficients on the lags of the causing variable is proportional to the long-run impact of that variable, the test can be viewed as a long-run Granger causality test. Importantly, the sum of the coefficients also indicates the estimated direction of the (long-run) relationship, such that the test is associated with a clear direction in the causation.

Table 2 shows the results, where the first five rows in each sub-panel provide the Granger causality results. The first three rows show the sum of the coefficients on the lags of the causing variable, along with the corresponding χ^2 statistic and p-value. The fourth and fifth rows in each sub-panel provide the results of the standard Granger causality test, namely that all of the coefficients on the lags of the causing variable are jointly equal to zero.¹⁷ The top panel shows tests of whether AT causes triangular arbitrage opportunities, whereas the bottom panel shows tests of whether triangular arbitrage opportunities cause algorithmic trading.

We first note that the relative presence of AT in the market (*VAT*) one minute has a statistically significant negative effect on triangular arbitrage the next minute: Algorithmic trading Granger causes a reduction in triangular arbitrage opportunities.¹⁸ There is also strong evidence, as seen from the lower half of the table, that an increase in triangular arbitrage opportunities Granger causes an increase in algorithmic trading activity. These relationships are seen even more clearly (judging by the size of the test statistics) when focusing on the taking activity of computers (*VCt* and *OF Ct*), where there is a very strong dual Granger causality between these measures of AT activity and triangular arbitrage. The making activity of computers (*VCm*), on the other hand, does not seem to be impacted by triangular arbitrage opportunities in a Granger-causal sense, and computer making actually Granger causes an *increase* in arbitrage opportunities (this result is, however, reversed in the contemporaneous identification, so we do not lend it much weight). One interpretation of these results is thus that increased AT activity this period (representing trading, of course, but probably also a higher level of monitoring) makes it less likely that prices will deviate from being arbitrage-free at the beginning of next period. Conversely, if prices are further away from arbitrage-free this period, arbitrage opportunities are more likely to exist at the beginning of the next period, and algorithmic traders are more likely to enter the market by taking liquidity on the relevant side to exploit those opportunities.

In summary, these findings suggest that algorithmic traders improve the informational efficiency of prices by taking advantage of arbitrage opportunities and making them disappear quickly, rather than by posting quotes that prevent these opportunities from occurring (as would be indicated by a strong negative effect of *VCm* on reducing triangular arbitrage opportunities). These results are also in line with the theoretical

¹⁷Since the standard Granger causality test is not associated with a clear direction in causation, we focus our discussion on the long-run test reported in the first through third rows of Table 2, which explicitly shows the direction of causation. Overall, the traditional Granger causality tests yield very similar results to the long-run tests in terms of statistical significance.

¹⁸The first lag of the causing variable tends to be the primary driver of the Granger causality relationship. Viewing the results as Granger causation from one minute to the next is therefore reasonable.

models of Biais, Foucault, and Moinas (2011), Martinez and Rosu (2011), and Oehmke (2009) and Kondor (2009).

The Granger causality results point to a strong dual causal relationship between AT (taking) activity and triangular arbitrage, such that increased AT causes a reduction in arbitrage opportunities and an increase in arbitrage opportunities causes an increase in AT activity. Although Granger causality tests are informative, they are based on the reduced form of the VAR, and do not explicitly identify the *contemporaneous* causal economic relationships in the model. We therefore also estimate the structural version of the model, using a version of the heteroskedasticity identification approach developed by Rigobon (2003) and Rigobon and Sack (2003, 2004). The basic idea of this identification scheme is that heteroskedasticity in the error terms can be used to identify simultaneous equation systems. If there are two distinct variance regimes for the error terms, this is sufficient for identification of the simultaneous structural VAR system defined in equation (1). In particular, if the covariance matrices under the two regimes are not proportional to each other, this is a sufficient condition for identification (Rigobon (2003)).

We rely on the observation that the variance of the different measures of AT participation in the market that we use in our analysis changes over time.¹⁹ To capitalize on this fact, we split our sample into two equal-sized sub-samples, simply the first and second halves of the sample period. Although the variances of the shocks are surely not constant within these two sub-samples, this is not crucial for the identification mechanism to work, as long as there is a clear distinction in (average) variance across the two sub-samples, as discussed in detail in Rigobon (2003). Rather, the crucial identifying assumption, as Stock (2010) points out, is that the structural parameters determining the contemporaneous impact between the variables (i.e., A in equation (1) above) are constant across the two variance regimes.²⁰ This is not an innocuous assumption, although it seems reasonable in our context as we have no reason to suspect (and certainly no evidence) that the underlying structural impact of the variables in our model changed over our sample period. Appendix A3 describes the mechanics of the actual estimation, which is performed via GMM, and also provides a more extensive discussion of the method. The Appendix also lists estimates of the covariance matrices across the two different regimes (sub-samples). The estimates show that there is strong heteroskedasticity between the first and second halves of the sample, and that the shift in the covariance matrices between the two regimes is not proportional, which is the key identifying condition mentioned above. Furthermore, we are reassured by the fact that in the vast majority of the statistically significant cases, the heteroskedasticity identification

¹⁹The variance of VAT , VCt , VCm , and $OFCT$ increase over time as AT participation increases, while the variance of the trade correlation measure, R , decreases over the sample period. Appendix A3 describes these results in detail.

²⁰Stock (2010) provides an overview of the heteroskedasticity identification method, among other methods, in his discussion of new “robust tools for inference,” classifying the heteroskedasticity identification method as a “quasi-experiment” approach. Some examples of recent studies that have used this method are Wright (2012), Erhmann, Fratzscher, and Rigobon (2011), Lanne and Lutkepohl (2008), and Primiceri (2005).

results are confirmed by the Granger causality analysis.

Estimates of the relevant contemporaneous structural parameters in equation (1) are shown in the sixth row of each sub-panel in Table 2, with Newey-West standard errors given in parentheses below. The eighth row in the table, shown in bold, provides a measure of the economic magnitude of the coefficients, obtained by multiplying the estimated contemporaneous effect for a given variable by its standard deviation.²¹ The magnitude of these estimates can be compared directly across the different measures of AT.

Overall, the contemporaneous effects are generally in line with those found in the Granger causality tests (with the exception of the sign of the impact of VCm , which switches). An increase in the presence of AT activity has a contemporaneous negative effect on triangular arbitrage opportunities, and an increase in triangular arbitrage simultaneously causes an increase in AT activity. The making activity (VCm) of algorithmic traders is now also associated with a reduction in arbitrage opportunities but, judging by the economic magnitude of the coefficients, $OF Ct$ and $\ln(R)$ have by far the biggest impact on arbitrage opportunities. In other words, high relative levels of computer order flow and high correlation of algorithmic strategies result in the largest reductions of this type of informational inefficiency. The results are not unexpected, as, together, they likely indicate that the correlated taking activity of computers in a similar direction plays an important role in extinguishing triangular arbitrage opportunities. It is also interesting to note that the economic impact of $\ln(R)$ is even higher than that of $OF Ct$. Since a high value of $\ln(R)$ indicates that computers trade predominantly with humans, this is consistent with the view advanced by Biais, Foucault, and Moinas (2011) and Martinez and Rosu (2011)—namely, that AT contributes positively to price discovery, but that it may also increase the adverse selection costs of slower traders.

In terms of the actual absolute level of economic significance, the estimated relationships are fairly large. For instance, consider the estimated contemporaneous impact of relative computer taking activity in the same direction ($OF Ct$) on triangular arbitrage. The contemporaneous effect is roughly -0.03 , and the scaled economic effect is roughly -0.5 . AT activity (including $OF Ct$) is measured in percent and triangular arbitrage is measured as the number of seconds in a minute with an arbitrage opportunity. A one percentage point increase in the directional intensity of computer trading ($OF Ct$) would thus, on average, reduce the number of seconds with a triangular arbitrage in a given minute by 0.03. This sounds like a small effect, but the (VAR residual) standard deviation of $OF Ct$ is around 17 percent, which suggests that a typical one-standard deviation move in $OF Ct$ results in a decrease of about 0.5 seconds of arbitrage opportunities in each minute; this is the estimated economic effect shown in bold in the table. As seen in the top panel of Figure 3, there are typically, in any given minute, only between 1 and 2 seconds with an arbitrage opportunity (that

²¹In particular, the coefficient for a given variable is scaled by the standard deviation of the VAR residuals for that variable, capturing the size of a typical innovation to the variable.

is, between 1.5 and 3 percent of the time), so a reduction of 0.5 seconds is quite sizeable. A one-standard deviation increase in “computer correlation” (i.e., in $\ln(R)$) causes an even larger estimated effect, with a resulting average reduction in arbitrage opportunities of 1.5 seconds per minute. A typical increase in overall AT participation (VAT) leads to 0.2 less seconds of triangular arbitrage opportunities each minute. Going the other direction, the causal impact of triangular arbitrage opportunities on AT activity is also fairly large. A typical increase in triangular arbitrage opportunities results in an approximately 4 percent increase in computer taking intensity ($OFCt$) and a 2 percent increase in overall computer participation (VAT).

In summary, both the Granger causality tests and the heteroskedasticity identification approach point to the same conclusion: Algorithmic trading causes a reduction in the frequency of triangular arbitrage opportunities. The taking activity of algorithmic traders seems to be an important mechanism through which this is accomplished, that is by computers hitting existing quotes in the system, with most of these quotes posted by human traders. Intervals of highly correlated computer trading activity also lead to a reduction in arbitrage opportunities, presumably as computers react simultaneously and in the same fashion to the same price data. Finally, and conversely, the presence of triangular arbitrage opportunities causes an increase in algorithmic trading activity, presumably as computers dedicated to detect this type of opportunity “lay in wait” until they detect a profit opportunity and then begin to place trading instructions.²²

6 High-Frequency Return Autocorrelation and AT

The impact of algorithmic trading on the frequency of triangular arbitrage opportunities is, however, only one facet of how computers affect the price discovery process. More generally, one often-cited concern is that algorithmic trading may contribute to the temporary deviation of asset prices from their fundamental values, resulting in excess volatility, particularly at high frequencies. We thus investigate the effect algorithmic traders have on a more general measure of prices not being informationally efficient: the autocorrelation of high-frequency currency returns. Specifically, we study the impact of algorithmic trading on the absolute value of the first-order autocorrelation of 5-second returns.^{23,24} We plot in Figure 4 the 50-day moving average

²²The interpretation of our evidence is consistent with Ito et al.’s (2012) finding that the probability of triangular arbitrage opportunities disappearing within a second has increased over time.

²³The random walk, or martingale hypothesis, was traditionally linked to the efficient market hypothesis (e.g., Samuelson (1965), Fama (1965), and Fama (1970), among others) —implying that efficient prices follow martingale processes and that returns exhibit no serial correlation, either positive (momentum) or negative (mean-reversion). We now know that predictability in prices is not inconsistent with rational equilibria and price efficiency when risk premia are changing. However it is highly unlikely that predictability in five-second returns reflects changing risk premia, and it seems reasonable to view high-frequency autocorrelation as an indicator of market (in-)efficiency.

²⁴Our choice of a 5-second frequency of returns is driven by a trade-off between sampling at a high enough frequency to estimate the effect algorithmic traders have on prices, but low enough to have sufficiently many transactions to avoid biasing the auto-correlations towards zero due to a high number of consecutive zero returns. Our results are qualitatively similar when we sample prices at the 1-second frequency for the most liquid currency pairs, euro-dollar and dollar-yen, but for the euro-yen the 1-second sampling frequency biases the return autocorrelations towards zero due to lower activity at the beginning of the sample.

of the absolute value of the autocorrelation of high-frequency (i.e., 5-second) returns estimated on a daily basis. Over time, the absolute value of autocorrelation in the least liquid currency pair, euro-yen, has declined drastically from an average value of 0.3 in 2003 to 0.03 in 2007, and the sharp decline in 2005 coincides with the growth of algorithmic trading participation. Again, of course, this is only suggestive evidence of the impact of AT. The decline in the absolute value of autocorrelation in dollar-yen is also notable, from 0.14 in 2003 to 0.04 in 2007, but there is no systematic change in the absolute value of autocorrelation in the most liquid currency pair traded, euro-dollar. The graphical evidence suggests that it is possible that algorithmic trading may have a greater effect on the informational efficiency of asset prices in less liquid markets.

In this section, we adopt the same identification approach that we used for triangular arbitrage opportunities, using a 5-minute frequency VAR (i.e., we estimate the autocorrelation of 5-second returns each 5-minute interval) that allows for both Granger causality tests as well as contemporaneous identification through the heteroskedasticity in the data across the sample period.²⁵

In the (structural) VAR analysis in this section, we follow a similar approach to the one used above for triangular arbitrage opportunities. However, since autocorrelation is measured separately for each currency pair (unlike the overall frequency of triangular arbitrage opportunities), we fit a separate VAR for each currency pair. Algorithmic trading activity is measured in the same five ways as before. We study the overall impact of the share of algorithmic trading activity in the market, VAT , and also that of the relative taking activity of computers, VCT , the relative making activity of computers, VCm , the relative presence of computer order flow, $OFCT$, and the degree of correlation in computer trading activity (and therefore strategies), $\ln(R)$.

Let $|AC_t^j|$ and AT_t^j be the five-minute measures of the absolute value of autocorrelation in 5-second returns and AT activity, respectively, in currency pair j , $j = 1, 2, 3$; as before, AT_t^j is used to represent either of the five measures of AT activity. The autocorrelations are expressed in percent. Define $Y_t^j = (|AC_t^j|, AT_t^j)$, and the structural form of the system is given by

$$A^j Y_t^j = \Phi^j(L) Y_t^j + \Lambda^j X_{t-1:t-20}^j + \Psi^j G_t + \epsilon_t^j. \quad (3)$$

A^j is the 2×2 matrix specifying the contemporaneous effects, normalized such that all diagonal elements are equal to 1. $X_{t-1:t-20}^j$ includes, for currency pair j , the sum of the volume of trade over the past 20 minutes and the volatility over the past 20 minutes, calculated as the sum of absolute returns over these 20

²⁵Our choice of estimating the autocorrelation over a 5-minute interval is driven by a trade-off between a large enough window with sufficient observations to estimate the autocorrelation (12 observations in a 1-minute window is too few observations) and small enough so that the Granger causality specification is sensible. We also perform a 99% Winsorising of the absolute values of the autocorrelation estimates (i.e., setting the largest 1% of the absolute autocorrelations equal to the 99th percentile of the absolute estimates).

minutes. Similar to our previous specification, G_t controls for secular trends and intra-daily patterns in the data by year-month dummy variables and half-hour dummy variables, respectively. The structural shocks ϵ_t^j are assumed to be independent of each other and serially uncorrelated at all leads and lags. The number of lags included in the VAR is set to 20. The corresponding reduced form equation is:

$$Y_t = (A^j)^{-1} \Phi^j(L) Y_t^j + (A^j)^{-1} \Lambda^j X_{t-1:t-20}^j + (A^j)^{-1} \Psi^j G_t + (A^j)^{-1} \epsilon_t^j \quad (4)$$

Table 3 shows the estimation results for both the reduced-form (equation (4)) and structural-form (equation (3)) VARs. The left hand panels show the results for tests of whether algorithmic trading has a causal impact on the absolute value of autocorrelations in 5-second returns, and the right hand panels show tests of whether the level of absolute autocorrelation has a causal impact on AT activity. The results are laid out in the same manner as for the triangular arbitrage case, with the only difference that each currency pair now corresponds to a separate VAR regression. In particular, the first five rows in each panel show the results from Granger causality tests based on the reduced form of the VAR in equation (4). The sixth, seventh, and eighth rows in each panel show the results from the contemporaneous identification of the structural form. Identification is again provided by the heteroskedasticity approach of Rigobon (2003) and Rigobon and Sack (2003, 2004). The estimation is performed in the same way as for the triangular arbitrage system, described in Appendix A3. Appendix A3 also shows and discusses the estimates of the covariance matrices for the two sub-samples used in the estimation.

The results in the right-hand panels do not reveal a clear pattern for the impact of higher absolute autocorrelation on algorithmic trading activity. In particular, the contemporaneous coefficients do not show a strong pattern of statistical significance and there is disagreement in some cases between the Granger causality results and the contemporaneous results.

The results in the left-hand panels of Table 3 are far more consistent across currencies and between the two estimation methods. First, the estimates clearly show that an increase in the share of AT participation (VAT) causes a decrease in the absolute autocorrelation of returns. As it did for triangular arbitrage, AT also appears to improve this measure of market efficiency. The Granger causality results show a role for both computer taking activity (VCt) and computer making activity (VCm) in the reduction of high-frequency autocorrelation, although the making results appear statistically stronger based on the significance and size of the test statistics. The results from the structural estimation show more clearly that the contemporaneous effect is much stronger for computer making than computer taking. This is seen both in the statistical significance of the estimates and the economic magnitude of the relationships (in bold). The evidence clearly indicates that, in contrast to the triangular arbitrage opportunity case, the improvement in the informational

efficiency of prices now predominantly comes from an increase in the trading activity of algorithmic traders who provide liquidity (i.e., those who post quotes which are hit), and not from an increase in the trading activity of algorithmic traders who hit quotes posted by others.

In terms of economic magnitude, the causal relationship between AT (making) activity and autocorrelation in returns seems somewhat less important than that between AT (taking) activity and triangular arbitrage, but still not insignificant. Recall that autocorrelation entered the VAR in a percentage format, and note that the scaled contemporaneous coefficients for VCm (in bold in the left-hand VCm panel of Table 3) are roughly between -0.3 and -0.4 . This suggests that a typical (one standard deviation) change in AT making activity, results in a drop of -0.3 to -0.4 percentage points in the absolute autocorrelation of returns. As seen in Figure 4, the average autocorrelations towards the end of the sample period are in the neighborhood of 5 percent for all three currency pairs. A third of a percentage point decrease in the autocorrelation is thus not a huge effect, but it does not appear irrelevant either.

Finally, it is also worth noting that in contrast to the triangular arbitrage case, the degree of correlation in computer trading strategies ($\ln(R)$) and the relative presence of computer order flow ($OF Ct$), both measures of commonality in algorithmic activity, do not appear to have a clear causal effect on autocorrelation.²⁶ As a matter of fact, when looking at the effects of these two variables, not one of the Granger or contemporaneous coefficients in our three currency pairs is statistically significant. In particular, there is no evidence that the high correlation of algorithmic strategies, which we showed earlier exists in this market, causes, on average, market inefficiencies, as measured by high-frequency autocorrelation.

7 Conclusion

Our paper studies the impact of algorithmic trading on the price discovery process in the global foreign exchange market using a long times series of high-frequency trading data in three major exchange rates. After describing the growth of algorithmic trading in this market since its introduction in 2003, we study its effect on two measures of price efficiency, the frequency of triangular arbitrage opportunities among our three exchange rates, and the presence of excess volatility, which we measure as the autocorrelation of high-frequency returns. Using a reduced-form and a structural VAR specification, we show that, on average, the presence of algorithmic trading in the market causes both a reduction in the frequency of arbitrage opportunities and a decrease in high-frequency excess volatility. Guided by the recent theoretical literature on the subject, we also study the mechanism by which algorithmic traders affect price discovery. We find evidence that algorithmic traders predominantly reduce arbitrage opportunities by acting on the quotes

²⁶As discussed in Appendix A3, there appears to be weak or no contemporaneous identification in the specification with $\ln(R)$ for the euro-yen currency pairs. All other specifications appear well identified.

posted by non-algorithmic traders. This result is consistent with the view that algorithmic trading improves informational efficiency by speeding up price discovery, but that, at the same time, it may increase the adverse selection costs to slower traders, as suggested by the theoretical models of Biais, Foucault, and Moinas (2011) and Martinez and Rosu (2011). In contrast, we show that the impact of algorithmic trading on excess volatility is more closely related to algorithmic trades that provide liquidity. In this case, consistent with Hoffman (2012), the improvement in the informational efficiency of prices may come from the fact that algorithmic quotes reflect new information more quickly. We also show evidence that is consistent with the actions and strategies of algorithmic traders being less diverse, more correlated, than those of non-algorithmic traders. But we find no evidence that, on average, a higher level of correlation leads to excess volatility.

Appendix

A1 Definition of Order Flow and Volume

The transactions data are broken down into categories specifying the “maker” and “taker” of the trades (human or computer), and the direction of the trades (buy or sell the base currency), for a total of eight different combinations. That is, the first transaction category may specify, say, the minute-by-minute volume of trade that results from a computer taker buying the base currency by “hitting” a quote posted by a human maker. We would record this activity as the human-computer buy volume, with the aggressor (taker) of the trade buying the base currency. The human-computer sell volume is defined analogously, as are the other six buy and sell volumes that arise from the remaining combinations of computers and humans acting as makers and takers.

From these eight types of buy and sell volumes, we can construct, for each minute, trading volume and order flow measures for each of the four possible pairs of human and computer makers and takers: human-maker/human-taker (HH), computer-maker/human-taker (CH), human-maker/computer-taker (HC), and computer-maker/computer-taker (CC). The *sum* of the buy and sell volumes for each pair gives the volume of trade attributable to that particular combination of maker and taker (denoted as $Vol(HH)$ or $Vol(HC)$, for example). The *sum* of the buy volume for each pair gives the volume of trade attributable to that particular combination of maker and taker, where the taker is buying (denoted as $Vol(HH^B)$ or $Vol(HC^B)$, for example). The *sum* of the sell volume for each pair gives the volume of trade attributable to that particular combination of maker and taker, where the taker is selling (denoted as $Vol(HH^S)$ or $Vol(HC^S)$, for example). The *difference* between the buy and sell volume for each pair gives the order flow attributable to that maker-taker combination (denoted as $OF(HH)$ or $OF(HC)$, for example). The sum of the four volumes, $Vol(HH + CH + HC + CC)$, gives the total volume of trade in the market. The sum of the four order flows, $OF(HH) + OF(CH) + OF(HC) + OF(CC)$, gives the total (market-wide) order flow.²⁷

Throughout the paper, we use the expression “volume” and “order flow” to refer both to the market-wide volume and order flow and to the volume and order flows from other possible decompositions, with the distinction clearly indicated. Importantly, the data allow us to consider volume and order flow broken down by the type of trader who initiated the trade, human-taker ($HH + CH$) and computer-taker ($HC + CC$); by the type of trader who provided liquidity, human-maker ($HH + HC$) and computer-maker ($CH + CC$); and by whether there was any computer participation ($HC + CH + CC$).

²⁷There is a very high correlation in this market between trading volume per unit of time and the number of transactions per unit of time, and the ratio between the two does not vary much over our sample. Order flow measures based on amounts transacted and those based on number of trades are therefore very similar.

A2 How Correlated Are Algorithmic Trades and Strategies?

In the benchmark model, there are H_m potential human-makers (the number of humans that are standing ready to provide liquidity), H_t potential human-takers, C_m potential computer-makers, and C_t potential computer-takers. For a given period of time, the probability of a computer providing liquidity to a trader is equal to $Prob(\text{computer} - \text{make}) = \frac{C_m}{C_m + H_m}$, which we label for simplicity as α_m , and the probability of a computer taking liquidity from the market is $Prob(\text{computer} - \text{take}) = \frac{C_t}{C_t + H_t} = \alpha_t$. The remaining makers and takers are humans, in proportions $(1 - \alpha_m)$ and $(1 - \alpha_t)$, respectively. Assuming that these events are independent, the probabilities of the four possible trades, human-maker/human-taker, computer-maker/human-taker, human-maker/computer-taker and computer-maker/computer taker, are:

$$\begin{aligned} Prob(HH) &= (1 - \alpha_m)(1 - \alpha_t) \\ Prob(HC) &= (1 - \alpha_m)\alpha_t \\ Prob(CH) &= \alpha_m(1 - \alpha_t) \\ Prob(CC) &= \alpha_m\alpha_t. \end{aligned}$$

These probabilities yield the following identity,

$$Prob(HH) \times Prob(CC) \equiv Prob(HC) \times Prob(CH),$$

which can be re-written as,

$$\frac{Prob(HH)}{Prob(CH)} \equiv \frac{Prob(HC)}{Prob(CC)}.$$

We label the first ratio, $RH \equiv \frac{Prob(HH)}{Prob(CH)}$, the “human-taker” ratio and the second ratio, $RC \equiv \frac{Prob(HC)}{Prob(CC)}$, the “computer-taker” ratio. In a world with more human traders (both makers and takers) than computer traders, each of these ratios will be greater than one, because $Prob(HH) > Prob(CH)$ and $Prob(HC) > Prob(CC)$; i.e., computers take liquidity more from humans than from other computers, and humans take liquidity more from humans than from computers. However, under the baseline assumptions of our random-matching model, the identity shown above states that the ratio of ratios, $R \equiv \frac{RC}{RH}$, will be equal to one. In other words, humans will take liquidity from other humans in a similar proportion that computers take liquidity from humans. Observing $R \neq 1$ therefore implies deviations from random matching. In particular, if computers trade less with each other, and more with humans, than expected, RC becomes larger than under random matching. By market clearing, this also implies that humans must trade less with each other, and more with computers, than expected, such that RH becomes smaller than under random matching.

Therefore, R must be greater than one in this case. The economic interpretation of $R > 1$ is discussed in detail in the main text.

To explicitly take into account the sign of trades, we amend the benchmark model as follows: we assume that the probability of the taker buying an asset is α_B and the probability of the taker selling is $1 - \alpha_B$. We can then write the probability of the following eight events (assuming each event is independent):

$$\begin{aligned}
\text{Prob}(HH^B) &= (1 - \alpha_m)(1 - \alpha_t)\alpha_B \\
\text{Prob}(HC^B) &= (1 - \alpha_m)\alpha_t\alpha_B \\
\text{Prob}(CH^B) &= \alpha_m(1 - \alpha_t)\alpha_B \\
\text{Prob}(CC^B) &= \alpha_m\alpha_t\alpha_B \\
\text{Prob}(HH^S) &= (1 - \alpha_m)(1 - \alpha_t)(1 - \alpha_B) \\
\text{Prob}(HC^S) &= (1 - \alpha_m)\alpha_t(1 - \alpha_B) \\
\text{Prob}(CH^S) &= \alpha_m(1 - \alpha_t)(1 - \alpha_B) \\
\text{Prob}(CC^S) &= \alpha_m\alpha_t(1 - \alpha_B)
\end{aligned}$$

These probabilities yield the following identities,

$$\begin{aligned}
\text{Prob}(HH^B) \times \text{Prob}(CC^B) &\equiv \text{Prob}(HC^B) \times \text{Prob}(CH^B) \\
(1 - \alpha_m)(1 - \alpha_t)\alpha_B\alpha_m\alpha_t\alpha_B &\equiv (1 - \alpha_m)\alpha_t\alpha_B\alpha_m(1 - \alpha_t)\alpha_B \\
&\text{and} \\
\text{Prob}(HH^S) \times \text{Prob}(CC^S) &\equiv \text{Prob}(HC^S) \times \text{Prob}(CH^S) \\
(1 - \alpha_m)(1 - \alpha_t)(1 - \alpha_B)\alpha_m\alpha_t(1 - \alpha_B) &\equiv (1 - \alpha_m)\alpha_t(1 - \alpha_B)\alpha_m(1 - \alpha_t)(1 - \alpha_B)
\end{aligned}$$

which can be re-written as,

$$\begin{aligned}
\frac{\text{Prob}(HH^B)}{\text{Prob}(CH^B)} &\equiv \frac{\text{Prob}(HC^B)}{\text{Prob}(CC^B)} \\
&\text{and} \\
\frac{\text{Prob}(HH^S)}{\text{Prob}(CH^S)} &\equiv \frac{\text{Prob}(HC^S)}{\text{Prob}(CC^S)}
\end{aligned}$$

We label the ratios, $RH^B \equiv \frac{\text{Prob}(HH^B)}{\text{Prob}(CH^B)}$, the “human-taker-buyer” ratio, $RC^B \equiv \frac{\text{Prob}(HC^B)}{\text{Prob}(CC^B)}$, the “computer-taker-buyer” ratio, $RH^S \equiv \frac{\text{Prob}(HH^S)}{\text{Prob}(CH^S)}$, the “human-taker-seller” ratio, and $RC^S \equiv \frac{\text{Prob}(HC^S)}{\text{Prob}(CC^S)}$, the “computer-

taker-seller" ratio.

In a world with more human traders (both makers and takers) than computer traders, each of these ratios will be greater than one, because $Prob(HH^B) > Prob(CH^B)$, $Prob(HH^S) > Prob(CH^S)$, $Prob(HC^B) > Prob(CC^B)$, and $Prob(HC^S) > Prob(CC^S)$. That is, computers take liquidity more from humans than from other computers, and humans take liquidity more from humans than from computers. However, under the baseline assumptions of our random-matching model, the identity shown above states that the ratio of ratios, $R^B \equiv \frac{RC^B}{RH^B}$, will be equal to one, and that $R^S \equiv \frac{RC^S}{RH^S}$ will also be equal to one. Observed values for R^B or R^S that deviate from one again provide evidence against trading patterns following random matching, with the empirically relevant case of $R^S, R^B > 1$ discussed in the main text.

A3 Heteroskedasticity Identification

Methodology

Restate the structural equation,

$$AY_t = \Phi(L)Y_t + \Lambda X_{t-1:t-20} + \Psi G_t + \epsilon_t,$$

and the reduced form,

$$Y_t = A^{-1}\Phi(L)Y_t + A^{-1}\Lambda X_{t-1:t-20} + A^{-1}\Psi G_t + A^{-1}\epsilon_t.$$

Let Ω_s be the variance-covariance matrix of the reduced form errors in variance regime s , $s = 1, 2$, which can be directly estimated by $\hat{\Omega}_s = \frac{1}{T_s} \sum_{t \in s} u_t u_t'$, where u_t are the reduced form residuals. Let $\Omega_{\epsilon,s}$ be the diagonal variance-covariance matrix of the structural errors and the following moment conditions hold,

$$A\Omega_s A' = \Omega_{\epsilon,s}. \quad (5)$$

The parameters in A and $\Omega_{\epsilon,s}$ can then be estimated with GMM, using estimates of Ω_s from the reduced form equation. Identification is achieved as long as the covariance matrices constitute a system of equations that is linearly independent. In the case of two regimes, the system is exactly identified. Standard errors are calculated using the Newey-West estimator.

It is useful to also provide some additional intuitive discussion on the heteroskedasticity identification approach. As mentioned, the basic idea is that if the variance of the shocks in the system changes over time,

but the structural parameters linking the variables to each other (i.e., the parameters in A in the current notation) remain constant, the system may be identified. An intuitive way of thinking of the method is that the shift in variances provides that extra source of variation needed for identification in the presence of endogeneity. In that sense, the method is similar in concept to typical instrumental variables estimation and, as discussed in Rigobon and Sack (2004), the heteroskedasticity identification approach can indeed be interpreted as an instrumental variable method. Purely heuristically, one might think of the (non-proportional) shift in variances as a form of relevance condition and the condition that the structural relationship captured by A is constant across variance regimes as a form of validity condition.²⁸

As with the usual validity condition for instrumental variables, the stability of A is generally not testable. Thus, provided that there is sufficient change in the variances of the shocks, which is relatively straightforward to investigate, the key condition for the method to work is parameter stability in A . This is not a trivial condition to satisfy. In cases where there is a perceived “structural change”, it might be difficult to argue that the variance of the structural shocks changed but the structural relationships remained constant. In other cases, where the shift in variances does not seem immediately associated with a “structural change”, the stability of the structural relationships seems more easily defensible. Ideally, one can point to a specific exogenous event that is likely to have affected the variances of the shocks but not the structural parameters in A ; such events are, of course, rare. Our case, it would seem, is somewhere in the middle of these scenarios. There is not one specific exogenous event that is likely to have caused the observed heteroskedasticity in our data, but there is also no obvious structural change that is likely to have triggered a shift in structural relationships. There have been a number of events on EBS during our sample period that may have affected the variance of our AT measures, including technological changes that allowed for faster electronic monitoring and responses, and the fact that more and more non-bank participants gained access to the system via prime-brokerage. But these changes were gradual and we cannot point to exact dates when they began to have an important influence. Importantly, we do not believe that these changes have influenced the fundamental way in which trades impact price discovery, such that the assumption of a stable structural relationship seems reasonable.

As always, with issues of identification, it is near-impossible to provide completely watertight arguments in any given situation. A more agnostic interpretation is therefore to view the results as conditional on the structural relationship remaining constant. That is, provided the relationship is constant over time, the method delivers consistent estimates of the causal effects..

²⁸An instrumental variable is generally referred to as valid if it is exogenous and relevant if it is correlated with the endogenous regressor.

Covariance Matrix Estimates Across Sub-Samples

The heteroskedasticity identification approach requires that there exist two linearly independent variance regimes. In particular, this implies that the covariance matrices for the errors in the two regimes are not proportional to each other. Table A1 shows the estimates of the covariance matrix for the reduced form VAR residuals for $Y_t = (Arb_t, AT_t^{avg})$, across the first and second half of the sample. As is clear, there is a strong increase in the variance of algorithmic trading, measured either as VAT , VCT , VCm , or $OFCT$. In contrast, the variance in triangular arbitrage opportunities is almost constant across the two sub-samples and it is immediately clear that the change in the covariance matrix between the two sub samples is not proportional. The variance of $\ln(R)$, which is the fifth measure of algorithmic trading activity that we use, is somewhat higher in the first sub-sample relative to the second sub-sample, although the distinction is less sharp than for the other measures of algorithmic trading. Since $\ln(R)$ is only observed when there are non-zero trades of all types, there are many missing observations (see Footnote 12). During the minutes when $\ln(R)$ is not missing, there is a fairly sharp decrease in the variance of triangular arbitrage opportunities. The shift in the covariance matrix from the first sub-sample to the second is therefore not proportional for $(Arb_t, \ln(R)_t^{avg})$ either, although the identification does not appear as strong as for the other AT measures.

Table A2 shows the corresponding results for the VAR residuals from the specification with algorithmic trading and high-frequency autocorrelation, $Y_t^j = (|AC_t^j|, AT_t^j)$. This specification is estimated separately for each currency pair, and the table reports results by currency pairs and by measures of AT . The story is similar to that in Table A1, with strong evidence that the covariance matrix does *not* shift in a proportional manner from the first to the second sub-sample. For the euro-dollar, the variance of the high-frequency autocorrelation increases somewhat from the first to the second sub-sample, but nowhere near as sharply as for the VAT , VCT , VCm , and $OFCT$ measures of algorithmic trading and the shift in the covariance matrix is not close to proportional. For euro-yen and the dollar-yen, the variance in high-frequency autocorrelation actually decreases from the first to the second sub-sample, whereas the variance of VAT , VCT , VCm , and $OFCT$ all increase, so again there seems to be strong identification. The variance of $\ln(R)$ decreases for all three currency pairs, and does so proportionally with the variance in high-frequency autocorrelation for the euro-yen. There therefore seems to be weak or no identification for the euro-yen specification using $\ln(R)$. For the euro-dollar and dollar-yen, there is, however, still identification for the $\ln(R)$ specifications.

Table A1: Covariances from residuals of the bivariate VAR with triangular arbitrage and algorithmic trading activity.

The table shows the covariance matrices for the residuals from the reduced form VAR in equation (2), which is estimated separately for each of the following measures of algorithmic trading activity averaged across currency pairs: Overall AT participation (VAT), AT taking participation (VCT), AT making participation (VCM), Relative Computer Taking ($OFCT$), and the natural logarithm of AT trade correlation ($\ln(R)$). The VAR is estimated using 1-minute data for the full sample from 2003 to 2007, spanning 1067 days. The residuals are split up into two sub samples, covering the first 534 days and last 533 days, respectively, and the covariance matrices are calculated separately for these two sub samples. The final column in the table shows the ratios between the estimates from the second and the first sub samples.

	First half of sample	Second half of sample	Ratio (second/first)
	<i>VAT</i>		
$Var(Arb)$	11.52	11.07	0.96
$Var(VAT)$	119.26	237.65	1.99
$Cov(Arb, VAT)$	5.32	3.31	0.62
No of Observations	256246	255770	
	<i>VCT</i>		
$Var(Arb)$	11.52	11.07	0.96
$Var(VCT)$	76.14	223.00	2.93
$Cov(Arb, VCT)$	5.06	3.86	0.76
No of Observations	256246	255770	
	<i>VCM</i>		
$Var(Arb)$	11.52	11.07	0.96
$Var(VCM)$	51.79	151.30	2.92
$Cov(Arb, VCM)$	0.42	-0.01	-0.02
No of Observations	256246	255770	
	<i>OFCT</i>		
$Var(Arb)$	11.50	11.05	0.96
$Var(OFCT)$	196.80	360.28	1.83
$Cov(Arb, OFCT)$	9.68	4.18	0.43
No of Observations	255972	255716	
	$\ln(R)$		
$Var(Arb)$	16.05	10.61	0.66
$Var(\ln(R))$	1.35	1.05	0.78
$Cov(Arb, \ln(R))$	0.78	0.18	0.23
No of Observations	13670	163043	

Table A2: Covariances from residuals of the bivariate VAR with high-frequency autocorrelation and algorithmic trading activity.

The table shows the covariance matrices for the residuals from the reduced form VAR in equation (4), which is estimated separately for each currency pair and each of the following measures of algorithmic trading activity: Overall AT participation (VAT), AT taking participation (VCT), AT making participation (VCm), Relative Computer Taking ($OFCT$), and the natural logarithm of AT trade correlation ($\ln(R)$). The VAR is estimated using 5-minute data for the full sample from 2003 to 2007, spanning 1067 days. The residuals are split up into two sub samples, covering the first 534 days and last 533 days, respectively, and the covariance matrices are calculated separately for these two sub samples. The final column in the table shows the ratios between the estimates from the second and the first sub samples.

	First half of sample				Second half of sample				Ratio (second/first)	
	USD/EUR	JPY/USD	JPY/EUR		USD/EUR	JPY/USD	JPY/EUR		USD/EUR	JPY/EUR
<i>VAT</i>										
$Var(ac)$	111.98	136.27	180.68		127.15	118.82	125.22		1.14	0.87
$Var(VAT)$	12.34	50.39	210.03		91.66	160.78	258.44		7.43	3.19
$Cov(ac, VAT)$	-0.22	-1.06	2.01		-3.95	-3.70	-2.31		18.02	3.49
No of Observations	51264	51260	51019		51168	51167	51094			-1.15
<i>VCT</i>										
$Var(ac)$	111.97	136.30	180.70		127.20	118.85	125.22		1.14	0.87
$Var(VCT)$	4.84	23.11	143.71		66.35	131.13	277.61		13.72	5.67
$Cov(ac, VCT)$	0.03	-0.60	-1.70		-1.57	-1.16	-1.63		-57.73	1.94
No of Observations	51264	51260	51019		51168	51167	51094			0.96
<i>VCm</i>										
$Var(ac)$	111.97	136.28	180.67		127.16	118.83	125.21		1.14	0.87
$Var(VCm)$	6.32	25.92	96.48		52.81	101.01	203.51		8.35	3.90
$Cov(ac, VCm)$	-0.27	-0.65	3.32		-3.54	-3.63	-1.00		12.98	5.60
No of Observations	51264	51260	51019		51168	51167	51094			-0.30
<i>OFCT</i>										
$Var(ac)$	111.89	136.10	180.31		127.22	118.90	125.09		1.14	0.87
$Var(OFCT)$	252.23	386.55	587.54		620.48	664.80	791.08		2.46	1.72
$Cov(ac, OFCT)$	0.02	2.64	2.37		1.47	0.24	1.17		67.63	0.09
No of Observations	51095	50871	49935		51155	51121	50880			0.49
$\ln(R)$										
$Var(ac)$	99.47	111.24	140.61		124.50	114.62	117.84		1.25	1.03
$Var(\ln(R))$	0.84	1.05	1.24		0.54	0.76	1.05		0.65	0.73
$Cov(ac, \ln(R))$	-0.07	-0.09	-0.09		-0.08	-0.09	-0.07		1.16	0.97
No of Observations	6114	5968	6732		44723	42298	37493			0.78

References

- [1] Berger, D.W., A.P. Chaboud, S.V. Chernenko, E. Howorka, and J.H. Wright, 2008. Order Flow and Exchange Rate Dynamics in Electronic Brokerage System Data, *Journal of International Economics* 75, 93-109.
- [2] Biais, B., Foucault, T., and Moinas S., 2011, Equilibrium Algorithmic Trading, Working Paper, Toulouse School of Economics.
- [3] Biais, B., and P. Woolley, 2011, High Frequency Trading, Manuscript, Toulouse University, IDEI.
- [4] Boehmer, E., Fong, K. Y. L., Wu, J., 2012, Algorithmic Trading and Changes in Firms' Equity Capital, Working Paper.
- [5] Brogaard, J. A., Hendershott, T., and Riordan, R., 2012, Algorithmic Trading and Price Discovery, working paper, University of California at Berkeley.
- [6] DeLong, B., Shleifer, A., Summers, L., Waldmann, R.J., 1990, Positive Feedback Investment Strategies and Destabilizing Rational Speculation, *Journal of Finance*, 45, 379-395.
- [7] Ehrmann, M., Fratzscher, M. and Rigobon, R., 2011, Stocks, Bonds, Money Markets and Exchange Rates: Measuring International Financial Transmission, *Journal of Applied Econometrics*, 26, 948-974.
- [8] Fama, E., 1965, The Behavior of Stock Market Prices, *Journal of Business*, 38.
- [9] Fama, E., 1970, Efficient Capital Markets: A Review of Theory and Empirical Work, *Journal of Finance*, 25.
- [10] Foucault, T., 2012, Algorithmic Trading: Issues and Preliminary Evidence, Market Microstructure Confronting Many Viewpoints, John Wiley & Sons.
- [11] Foucault, T., and Menkveld, A., 2008, Competition for Order Flow and Smart Order Routing Systems, *Journal of Finance*, 63, 119-158.
- [12] Froot, K., Scharfstein, D. S., Stein, J.C., 1992, Herd on the Street: Informational Inefficiencies in a Market with Short-Term Speculation, *Journal of Finance*, 47, 1461-1484.
- [13] Hasbrouck, J., and Saar, G., 2012. Low-Latency Trading, Working Paper.
- [14] Hendershott, T., C.M. Jones, and A.J. Menkveld, 2011. Does Algorithmic Trading Improve Liquidity?, *Journal of Finance*, 66, 1-33.

- [15] Hendershott, T., and Riordan, R., 2012, Algorithmic Trading and the Market for Liquidity, *Journal of Financial and Quantitative Analysis*, forthcoming.
- [16] Hirschey, N., 2011, Do High Frequency Traders Anticipate Buying and Selling Pressure? University of Texas at Austin, Working Paper.
- [17] Hoffman, P., 2012, A Dynamic Limit Order Market with Fast and Slow Traders, Working Paper.
- [18] Ito, T., Yamada, K., Takayasu, M., and Takayasu, H., 2012, Free Lunch! Arbitrage Opportunities in the Foreign Exchange Markets, NBER Paper No. 18541.
- [19] Jovanovic, B., and A.J. Menkveld, 2011, Middlemen in Limit-Order Markets, Working Paper.
- [20] Khandani, A., and Lo, A., 2007, What Happened to the Quants in August 2007?, *Journal of Investment Management* 5, 5-54.
- [21] Khandani, A., and Lo, A., 2011, What Happened to the Quants in August 2007? Evidence from Factors and Transactions Data, *Journal of Financial Markets* 14, 1-46.
- [22] Kirilenko, A., A. S. Kyle, M. Samadi, and T. Tuzun, 2011, The Flash Crash: The Impact of High Frequency Trading on an Electronic Market, Working Paper.
- [23] Kondor, P., 2009, Risk in Dynamic Arbitrage: Price Effects of Convergence Trading, *Journal of Finance*, 64, 631-655.
- [24] Kozhan, R., and W. Wah Tham, 2012, Execution Risk in High-Frequency Arbitrage, *Management Science*, forthcoming.
- [25] Lanne, M., and Lutkepohl, H., 2008, Identifying Monetary Policy Shocks via Changes in Volatility, *Journal of Money, Credit and Banking*, 40, 1131-1149.
- [26] Martinez, V., and I. Roşu, 2011, High Frequency Traders, News and Volatility, Working Paper.
- [27] Menkveld, A., 2011, High Frequency Trading and the New-Market Makers, Working Paper.
- [28] Oehmke, M., 2009, Gradual Arbitrage, Working Paper, Columbia University.
- [29] Primiceri, G. E., 2005, Time Varying Structural Vector Autoregressions and Monetary Policy, *Review of Economic Studies*, 72, 821-852.
- [30] Rigobon, R., 2003, Identification Through Heteroskedasticity, *Review of Economics and Statistics*, 85, 777-792.

- [31] Rigobon, R., and Sack, B., 2003, Measuring the Reaction of Monetary Policy to the Stock Market, *Quarterly Journal of Economics*, 118, 639-669.
- [32] Rigobon, R., and Sack, B., 2004, The Impact of Monetary Policy on Asset Prices, *Journal of Monetary Economics*, 51, 1553-1557.
- [33] Samuelson, P. A., 1965, Proof that Properly Anticipated Prices Fluctuate Randomly, *Industrial Management Review*.
- [34] Stein, J., 2009, Presidential Address: Sophisticated Investors and Market Efficiency, *Journal of Finance*, 64, 1571-1548.
- [35] Stock, J., 2010, The Other Transformation in Econometric Practice: Robust Tools for Inference, *Journal of Economic Perspective*, 24, 83-94
- [36] Wright, J., 2012, What Does Monetary Policy do To Long-term Interest Rates at the Zero Lower Bound? Working Paper, Johns Hopkins University.

Table 1: Correlation among algorithmic trading strategies: R-measure.

The table reports estimates of the relative degree to which computers trade with each other compared to how much they trade with humans, based on the benchmark model described in the main text. In particular, we report mean estimates of the log of $R = RC/RH$, $R^S = RC^S/RH^S$, and $R^B = RC^B/RH^B$, measured at the 1-minute, 5-minute, and daily frequency, with standard errors shown in parentheses below the estimates. $\ln(R)$, $\ln(R^S)$, $\ln(R^B) > 0$ (R , R^S , $R^B > 1$) indicates that computers trade less with each other than random matching would predict. The fraction of observations where $\ln(R)$, $\ln(R^S)$, $\ln(R^B) > 0$ is also reported along with the number of non-missing observations and the total number of observations, at each frequency. The ***, **, and * represent a statistically significant deviation from zero at the 1, 5, and 10 percent level, respectively. The sample period is September 2003 to December 2007.

	1-min data			5-min data			Daily data		
	$\ln(R)$	$\ln(R^S)$	$\ln(R^B)$	$\ln(R)$	$\ln(R^S)$	$\ln(R^B)$	$\ln(R)$	$\ln(R^S)$	$\ln(R^B)$
Mean	0.2219***	0.0544***	0.0401***	0.3674***	0.2118***	0.2078***	0.5314***	0.4901***	0.4995***
(std. err.)	(0.0031)	(0.0042)	(0.0042)	(0.0036)	(0.0046)	(0.0047)	(0.0118)	(0.0122)	(0.0118)
Fraction of obs. > 0	0.599	0.527	0.522	0.723	0.627	0.626	0.990	0.975	0.974
No. of non-missing obs.	143421	89914	87555	52101	44310	43560	880	846	854
Total no. of obs.	512160	512160	512160	102432	102432	102432	1067	1067	1067
JPY/USD									
Mean	0.3106***	0.1536***	0.1515***	0.3925***	0.2120***	0.2113***	0.5849***	0.5758***	0.5401***
(std. err.)	(0.0037)	(0.0052)	(0.0052)	(0.0043)	(0.0054)	(0.0054)	(0.0131)	(0.0144)	(0.0134)
Fraction of obs. > 0	0.611	0.545	0.545	0.703	0.609	0.610	0.990	0.969	0.971
No. of non-missing obs.	114449	61023	61645	49906	39036	39362	953	938	922
Total no. of obs.	512160	512160	512160	102432	102432	102432	1067	1067	1067
JPY/EUR									
Mean	0.6987***	0.5310***	0.5198***	0.6873***	0.5842***	0.5838***	0.8130***	0.7733***	0.7380***
(std. err.)	(0.0049)	(0.0074)	(0.0076)	(0.0051)	(0.0066)	(0.0068)	(0.0173)	(0.0171)	(0.0164)
Fraction of obs. > 0	0.696	0.643	0.636	0.758	0.700	0.698	0.984	0.975	0.965
No. of non-missing obs.	71783	30567	29337	45846	31629	31010	987	951	942
Total no. of obs.	512160	512160	512160	102432	102432	102432	1067	1067	1067

Table 2: Triangular arbitrage and algorithmic trading.

We report tests of whether algorithmic trading activity has a causal impact on triangular arbitrage (top panels) and whether triangular arbitrage has a causal impact on algorithmic trading activity (bottom panels). Results are presented separately for each of the following measures of algorithmic trading activity averaged across currency pairs: Overall AT participation (*VAT*), AT taking participation (*VCt*), AT making participation (*VCm*), Relative Computer Taking (*OFCt*), and the natural logarithm of AT trade correlation ($\ln(R)$). All results are based on 1-minute data covering the full sample period from 2003 to 2007. The first five rows in each panel present the results from Granger causality tests, based on the reduced form VAR in equation (2). In particular, the first three rows in each panel report the sum of the lag-coefficients for the causing variable, and the corresponding Wald χ^2 -statistic and p-value for the null hypothesis that the sum of the coefficients is equal to zero, respectively. The fourth and fifth rows report the Wald χ^2 -statistic and p-value, respectively, for the null hypothesis that the coefficients on all lags of the causing variable are jointly equal to zero. The following row, labeled Contemp. coeff., presents the point estimate of the contemporaneous impact of the causing variable in the structural VAR in equation (1), based on the heteroskedasticity identification scheme described in the main text, with the Newey-West standard error presented below in parentheses. The contemporaneous impact scaled by the residual standard deviation, as a measure of economic magnitude, is reported in the second to last row. The last row in each panel shows the total number of observations available for estimation in the full sample. The ***, **, and * represent a statistically significant deviation from zero at the 1, 5, and 10 percent level, respectively. The rows in bold show the economic magnitude of the effect.

Tests of AT Causing Triangular Arbitrage				
	<i>VAT</i>	<i>VCt</i>	<i>VCm</i>	<i>OFCt</i>
Sum of coeffs. on <i>AT</i> lags	-0.0027***	-0.0061***	0.0049***	-0.0117***
χ^2_1 (Sum = 0)	7.7079***	25.2501***	14.4351***	142.4661***
p-value	0.0055	0.0000	0.0001	0.0000
χ^2_{20} (All coeffs. on <i>AT</i> lags = 0)	91.4066***	149.5884***	46.5108***	317.3994***
p-value	0.0000	0.0000	0.0007	0.0037
Contemp. coeff.	-0.0145***	-0.0067**	-0.0040***	-0.0288***
(std. err.)	(0.0040)	(0.0027)	(0.0011)	(0.0055)
Contemp. coeff. $\times \sigma_{AT}$	-0.1932***	-0.0813**	-0.0406***	-0.4813***
No. of obs.	512016	512016	512016	511688
				176713
Tests of Triangular Arbitrage Causing <i>AT</i>				
	<i>VAT</i>	<i>VCt</i>	<i>VCm</i>	<i>OFCt</i>
Sum of coeffs. on <i>arb</i> lags	0.0140***	0.0219***	-0.0022	0.0345***
χ^2_1 (Sum = 0)	19.5814***	56.9040***	0.8233	75.9510***
p-value	0.0000	0.0000	0.3642	0.0000
χ^2_{20} (All coeffs. on <i>arb</i> lags = 0)	83.0479***	118.9288***	31.0039*	459.3056***
p-value	0.0000	0.0000	0.0551	0.0000
Contemp. coeff.	0.6072***	0.4817***	0.0545***	1.3037***
(std. err.)	(0.0591)	(0.0331)	(0.0089)	(0.1241)
Contemp. coeff. $\times \sigma_{arb}$	2.0405***	1.6185***	0.1832	4.3776***
No. of obs.	512016	512016	512016	511688
				176713

Table 3: Autocorrelation of high-frequency returns and algorithmic trading.

We report tests of whether algorithmic trading activity has a causal impact on autocorrelation (left hand panels) and whether autocorrelation has a causal impact on algorithmic trading activity (right hand panels). Results are presented separately for each currency pair and for each of the following measures of algorithmic trading activity averaged across currency pairs: Overall AT participation (*VAT*), AT taking participation (*VCT*), AT making participation (*VCM*), Relative Computer Taking (*OFCt*), and the natural logarithm of AT trade correlation ($\ln(R)$). All results are based on 1-minute data covering the full sample period from 2003 to 2007. The first five rows in each panel present the results from Granger causality tests, based on the reduced form VAR in equation (4). In particular, the first three rows in each panel report the sum of the lag-coefficients for the causing variable, and the corresponding Wald χ^2 -statistic and p-value for the null hypothesis that the sum of the coefficients is equal to zero, respectively. The fourth and fifth rows report the Wald χ^2 -statistic and p-value, respectively, for the null hypothesis that the coefficients on all lags of the causing variable are jointly equal to zero. The following row, labeled Contemp. coeff., presents the point estimate of the contemporaneous impact of the causing variable in the structural VAR in equation (3), based on the heteroskedasticity identification scheme described in the main text, with the Newey-West standard error presented below in parentheses. The contemporaneous impact scaled by the residual standard deviation, as a measure of economic magnitude, is reported in the second to last row. The last row in each panel shows the total number of observations available for estimation in the full sample. The ***, **, and * represent a statistically significant deviation from zero at the 1, 5, and 10 percent level, respectively. The rows in bold show the economic magnitude of the effect.

USD/EUR			JPY/USD			JPY/EUR		
Tests of <i>VAT</i> Causing Autocorrelation			Tests of Autocorrelation Causing <i>VAT</i>					
Sum of coeffs. on <i>VAT</i> lags	-0.03904***	-0.02562***	Sum of coeffs. on <i>ac</i> lags	-0.00689*	-0.01029*			
χ^2 (Sum = 0)	33.7507***	25.2649***	χ^2 (Sum = 0)	3.0155**	3.4939**			
p-value	0.0000	0.0000	p-value	0.0825	0.0616			
χ^2_0 (All coeffs. on <i>VAT</i> lags = 0)	43.0591***	45.5455***	χ^2_0 (All coeffs. on <i>ac</i> lags = 0)	12.8762**	4.0260			
p-value	0.0000	0.0000	p-value	0.0119	0.4025			
Contemp. coeff.	-0.04770***	-0.02377***	Contemp. coeff.	0.00330*	0.00100			
(std. err.)	(0.00703)	(0.00622)	(std. err.)	(0.00188)	(0.00427)			
Contemp. coeff. $\times \sigma_{VAT}$	-0.34388***	-0.24414***	Contemp. coeff. $\times \sigma_{ac}$	0.03607*	0.01129			
No. of obs.	102432	102427	No. of obs.	102432	102427			
Tests of <i>VCT</i> Causing Autocorrelation			Tests of Autocorrelation Causing <i>VCT</i>					
Sum of coeffs. on <i>VCT</i> lags	-0.04325***	-0.02187***	Sum of coeffs. on <i>ac</i> lags	-0.00019	-0.00123			
χ^2 (Sum = 0)	23.0192***	11.8863***	χ^2 (Sum = 0)	0.0033	0.0688			
p-value	0.0000	0.0006	p-value	0.9545	0.7931			
χ^2_0 (All coeffs. on <i>VCT</i> lags = 0)	24.7961***	20.8758***	χ^2_0 (All coeffs. on <i>ac</i> lags = 0)	2.6777	1.7370			
p-value	0.0001	0.0003	p-value	0.6131	0.7840			
Contemp. coeff.	-0.02630***	-0.00577	Contemp. coeff.	0.00138	-0.00343			
(std. err.)	(0.00752)	(0.00559)	(std. err.)	(0.00105)	(0.00247)			
Contemp. coeff. $\times \sigma_{VCT}$	-0.15686***	-0.05066	Contemp. coeff. $\times \sigma_{ac}$	0.01509	-0.03875			
No. of obs.	102432	102427	No. of obs.	102432	102427			

Table 3: High-frequency autocorrelation and algorithmic trading: (cont.)

	USD/EUR	JPY/USD	JPY/EUR	USD/EUR	JPY/USD	JPY/EUR
Tests of VCm Causing Autocorrelation						
Sum of coeffs. on VCm lags	-0.04692***	-0.03620***	-0.02146***	Sum of coeffs. on ac lags	-0.00917***	-0.01358***
χ^2_1 (Sum = 0)	29.7277***	28.8421***	17.8840***	χ^2_1 (Sum = 0)	9.3691	10.1188
p-value	0.0000	0.0000	0.0000	p-value	0.0022	0.0015
χ^2_{20} (All coeffs. on VCm lags = 0)	44.1685***	38.8638***	18.9016***	χ^2_{20} (All coeffs. on ac lags = 0)	33.8031***	11.0068**
p-value	0.0000	0.0000	0.0008	p-value	0.0000	0.0265
Contemp. coeff.	-0.07073***	-0.03907***	-0.02409***	Contemp. coeff.	0.00156	0.00268
(std. err.)	(0.00902)	(0.00711)	(0.00622)	(std. err.)	(0.00137)	(0.00269)
Contemp. coeff. $\times \sigma_{VCm}$	-0.38444***	-0.31120***	-0.29510***	Contemp. coeff. $\times \sigma_{ac}$	0.01707	0.03027
No. of obs.	102432	102427	102113	No. of obs.	102432	102427
Tests of $OFCt$ Causing Autocorrelation						
Sum of coeffs. on $OFCt$ lags	-0.00481	-0.00154	0.00013	Sum of coeffs. on ac lags	-0.02571**	-0.01388
χ^2_1 (Sum = 0)	2.3004	0.2637	0.0021	χ^2_1 (Sum = 0)	5.0050**	1.2702
p-value	0.1293	0.6076	0.9638	p-value	0.0253	0.2597
χ^2_{20} (All coeffs. on $OFCt$ lags = 0)	2.8266	1.8544	1.3831	χ^2_{20} (All coeffs. on ac lags = 0)	8.5794*	4.7680
p-value	0.5872	0.7625	0.8471	p-value	0.0725	0.3119
Contemp. coeff.	0.00434	-0.00631	-0.00124	Contemp. coeff.	-0.00958	0.03736**
(std. err.)	(0.00455)	(0.00463)	(0.00464)	(std. err.)	(0.01482)	(0.01857)
Contemp. coeff. $\times \sigma_{OFCt}$	0.09057	-0.14479	-0.03253	Contemp. coeff. $\times \sigma_{ac}$	-0.10473	0.42178**
No. of obs.	102250	101992	100815	No. of obs.	102250	101992
Tests of $\ln(R)$ Causing Autocorrelation						
Sum of coeffs. on $\ln(R)$ lags	-0.03022	0.00984	0.06147	Sum of coeffs. on ac lags	0.00056	-0.00050
χ^2_1 (Sum = 0)	0.06553	0.00931	0.40488	χ^2_1 (Sum = 0)	0.91886	0.46341
p-value	0.79795	0.92312	0.52458	p-value	0.33778	0.49604
χ^2_{20} (All coeffs. on $\ln(R)$ lags = 0)	3.09457	2.66549	0.96692	χ^2_{20} (All coeffs. on ac lags = 0)	5.05363	1.69574
p-value	0.54213	0.61527	0.91477	p-value	0.28184	0.79149
Contemp. coeff.	-0.01286	-0.01857	0.00017	Contemp. coeff.	-0.00058	-0.00066
(std. err.)	(0.30686)	(0.49029)	(13.18378)	(std. err.)	(0.00146)	(0.00340)
Contemp. coeff. $\times \sigma_{\ln(R)}$	-0.00978	-0.01656	0.00017	Contemp. coeff. $\times \sigma_{ac}$	-0.00643	-0.00706
No. of obs.	50837	48266	44225	No. of obs.	50837	48266

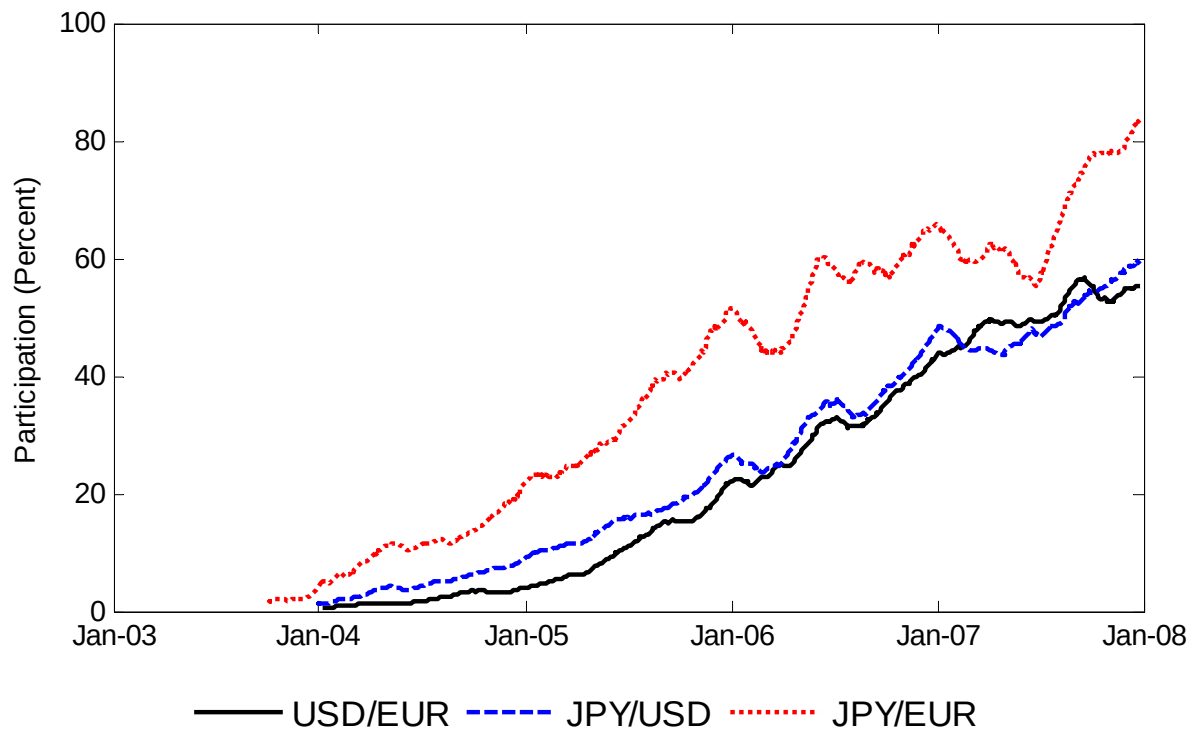


Figure 1: 50-day moving averages of VAT, the percent of total volume with at least one algorithmic counterparty.

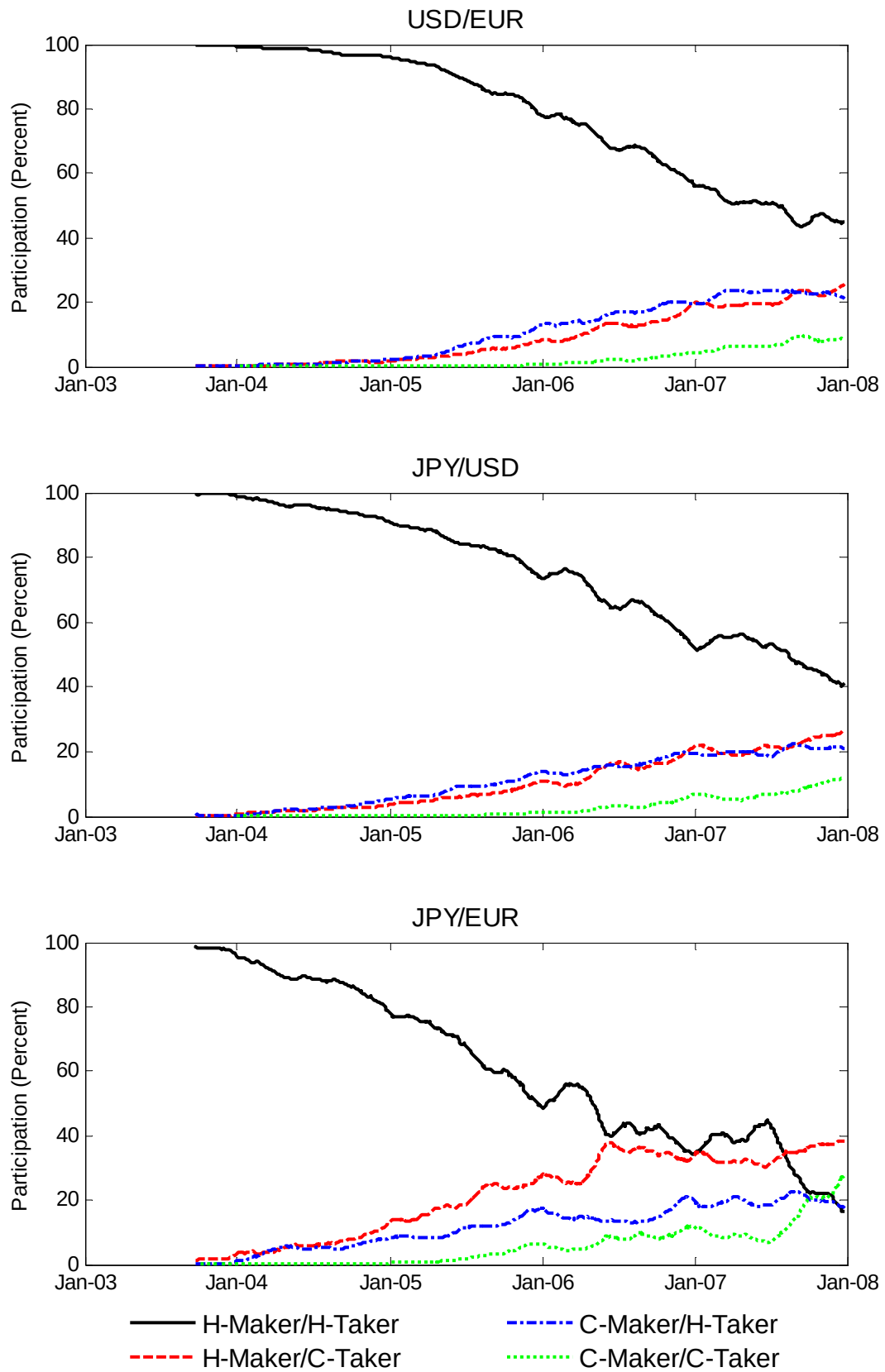


Figure 2: 50-day moving averages of the percent of total volume broken down into four maker-taker pairs.

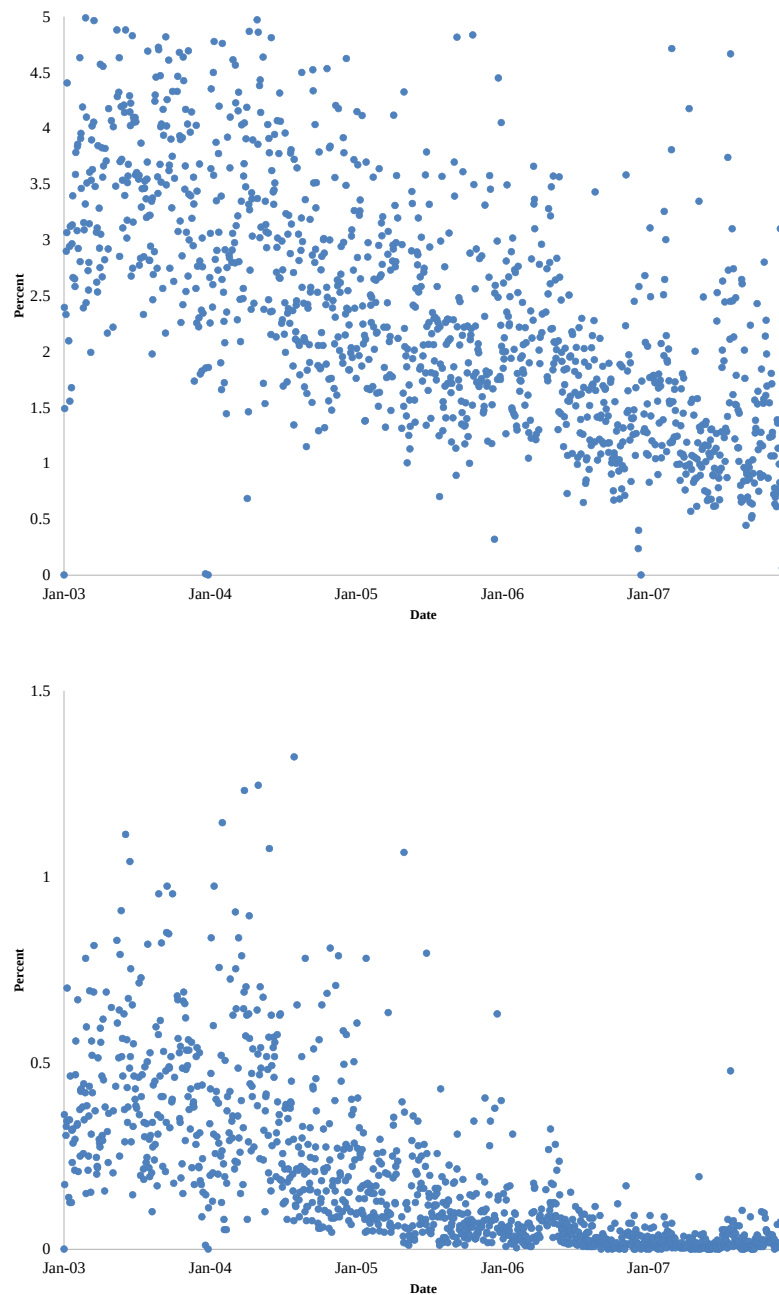


Figure 3: Percent of seconds with a triangular arbitrage opportunity with a profit strictly greater than 0 basis point (top panel) and 1 basis point (bottom panel).

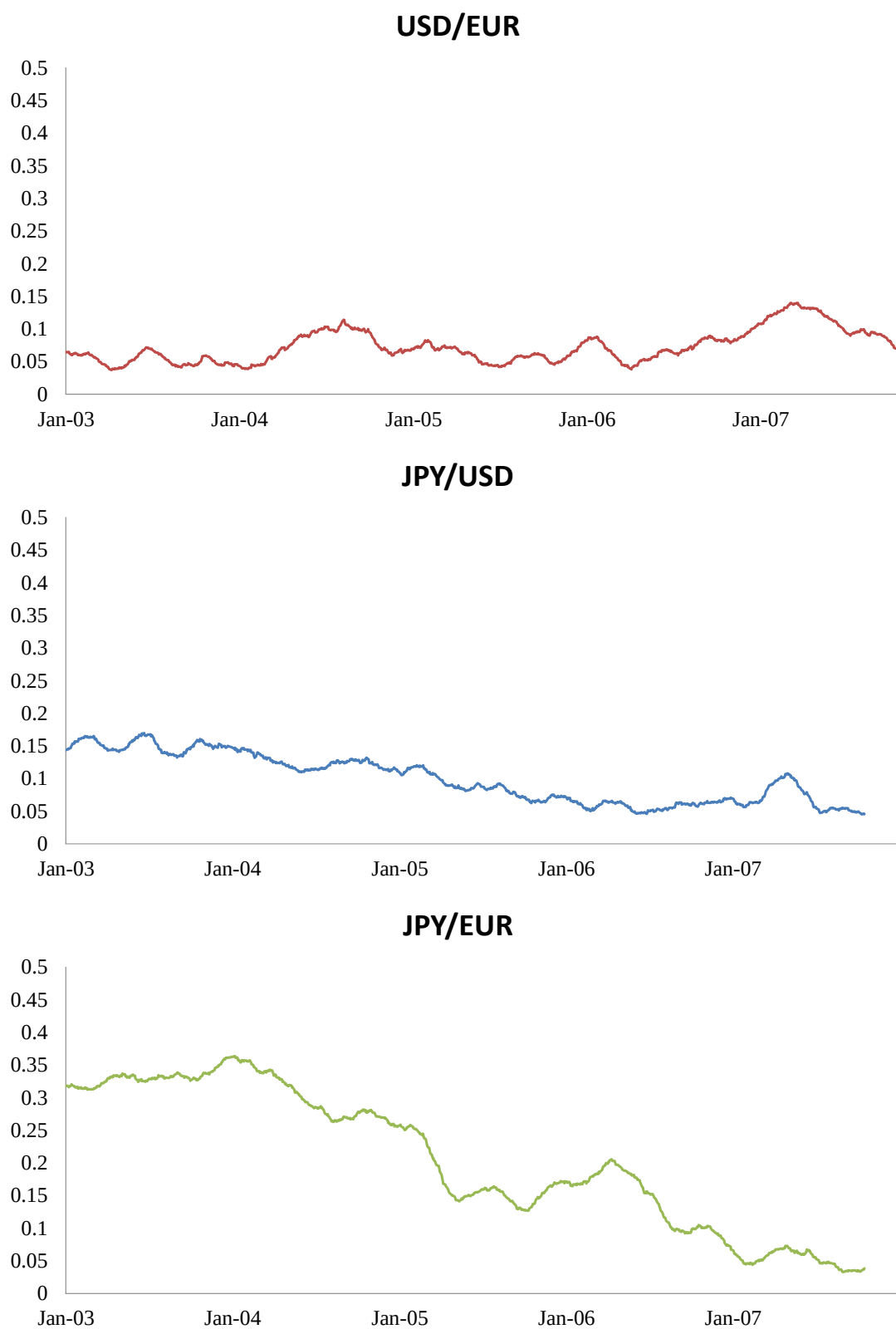


Figure 4: 50-day moving averages of absolute value of 5-second return serial autocorrelation estimated each day.